

CIRCUIT THEORY II - R4053 - 2025	
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T.P. No. 2: Modern Filter Theory	
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Observations:	

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Exercise #1

The low-pass filter (image 1) should present a Butterworth response with a bandwidth of -3dB at 1KHz. Determine the value of L and C knowing that the load is 1 Kohm. Verify the filter specifications using simulators. Suggestion: design the filter with normalized frequency and impedance values and then de-normalize to calculate the required filter value

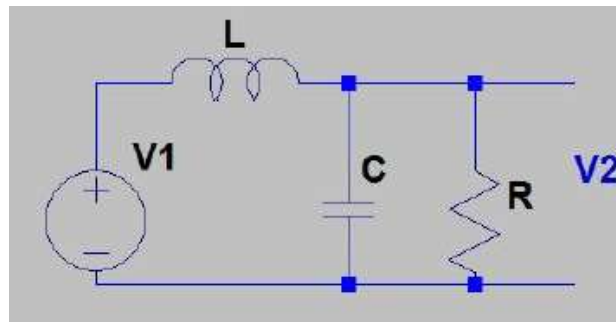


Imagen 1

Low-pass filter circuit to be designed.

Ej 1

Resposta Butterworth, BW -3dB a 1KHz
Carga de 1K Ω , valores normalizados en freq e impedancia y luego desnormalizar

$\rightarrow$ teniendo la transferencia del circuito

$$H(s) = \frac{V_2}{V_1} = \frac{R}{sL + \frac{1}{sC} + R}$$

$$H(s) = \frac{1}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

ta sen 2

$\omega_0 = \frac{1}{RC}$ $\omega_0 = \frac{1}{LC}$

$Q = \omega_0 R C = R \sqrt{\frac{C}{L}}$

$Q = \frac{1}{\sqrt{LC}} RC = R \sqrt{\frac{C}{L}}$

$\rightarrow$ A partir de la expresión genérica de Butterworth orden 2

$H(s) = \frac{1}{s^2 + \sqrt{2}s + 1}$ en $\phi = \frac{\pi}{4}$ y $\cos \phi = \frac{\sqrt{2}}{2}$

$H(s) = \frac{1}{s^2 + \frac{\sqrt{2}}{Q}s + 1}$ $\frac{1}{Q} = \sqrt{2} \rightarrow Q = \frac{1}{\sqrt{2}} = 0.707$

$\frac{1}{Q} = \frac{1}{\sqrt{2}} \rightarrow \frac{1}{RC} = \frac{1}{\sqrt{2}} \rightarrow RC = 0.707$

$\rightarrow$ Seguimos mezclando ecuaciones para obtener el resto de valores

$Q = R \sqrt{\frac{C}{L}}$ con R normalizado = 1 $\frac{1}{\sqrt{2}} = 1 \sqrt{\frac{C}{L}} \Rightarrow \frac{1}{2} = \frac{C}{L} \Rightarrow L = 2C = 1mH$

$\rightarrow$ Normas de normalización para impedancias y frecuencias

$Z_N = 1K$ $R = R_N \cdot Z_N = 1 \cdot 1K = 1K$

$f_N = 2\pi \cdot 1K$ $L = L_N \cdot Z_N = 1 \cdot 1K = 1K$

$C = C_N \cdot \frac{1}{Z_N \cdot f_N} = 1 \cdot \frac{1}{1K \cdot 2\pi \cdot 1K} = 15.9nF$

verifico $f_{cutoff} = 1000, 23dB$

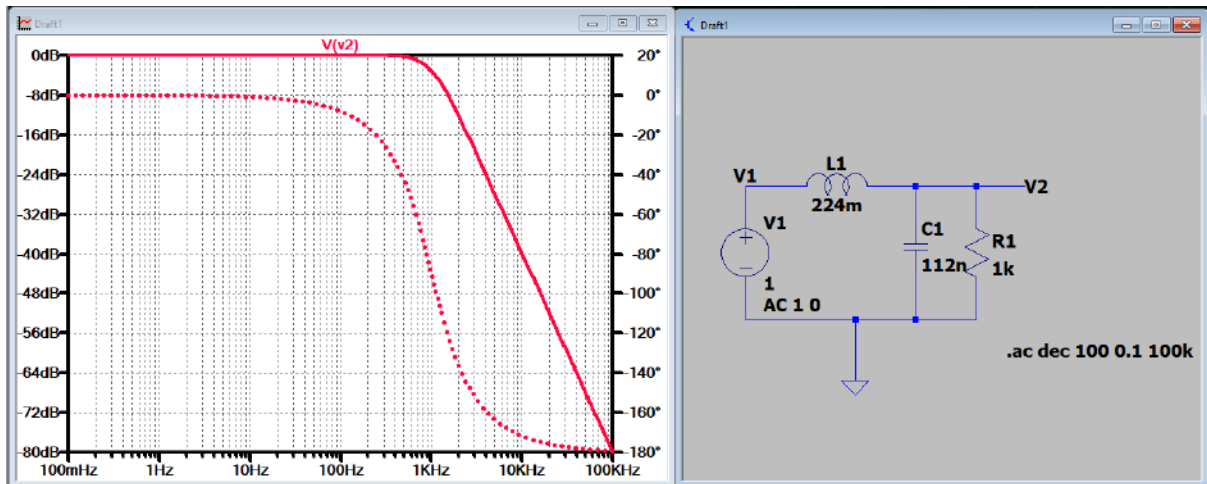


Imagen 2

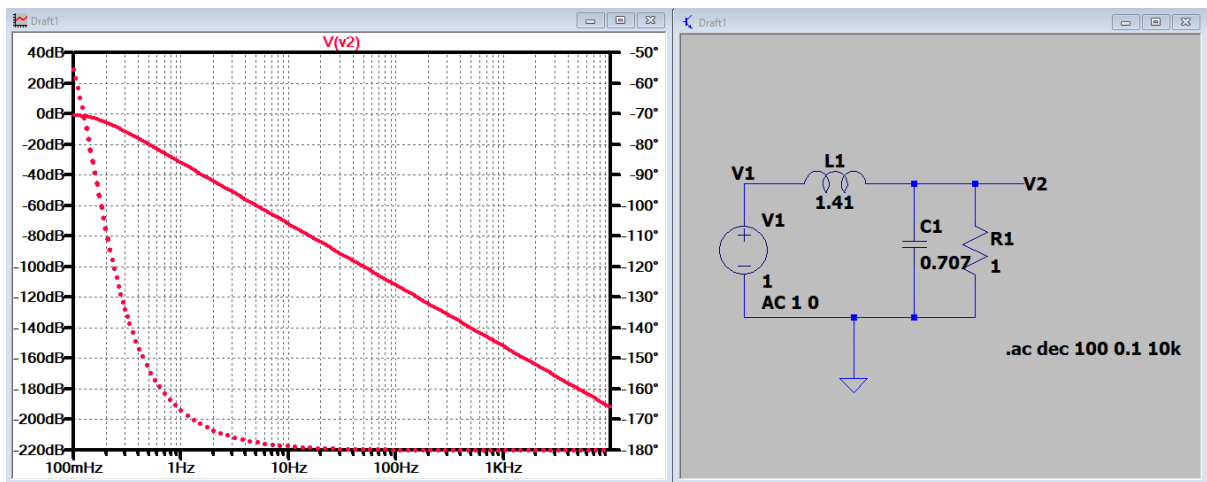
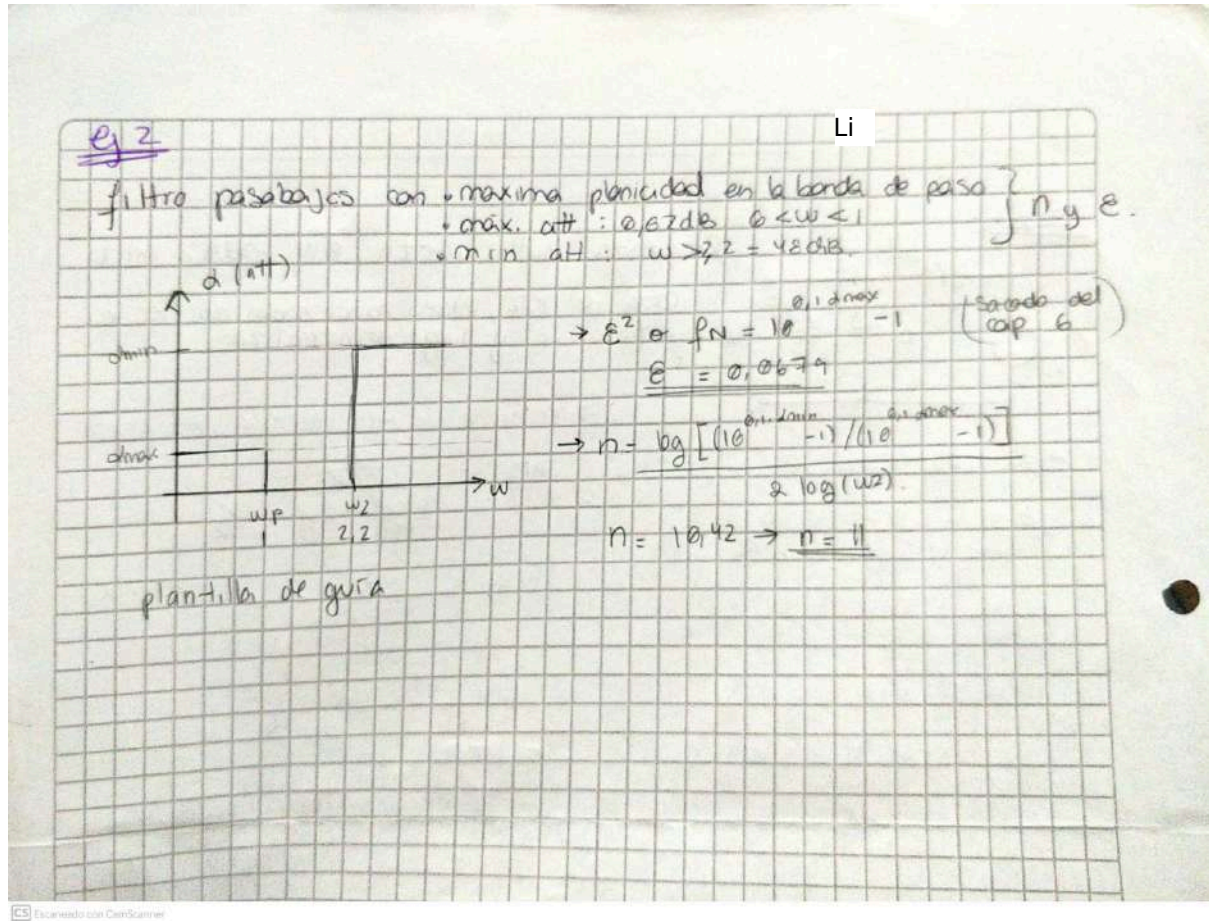


Imagen 3

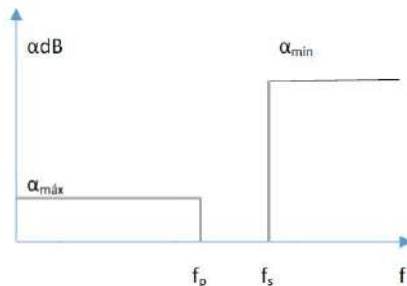
Exercise #2

It is desired to design a low-pass filter with the following characteristics. ● Maximum flatness in the passband ● Maximum attenuation: 0.02 dB for $0 < \omega < 1$ ● Minimum attenuation for $\omega > 2.2 = 48$ dB The parameters n and ϵ are to be obtained



Exercise #3

Determine the filter order for maximum flatness - Determine the location of the poles and the Q associated with them - Design the filters if a 20dB gain in the passband is desired, using Sallen Key structures for 3.1 and 3.3 and MFB for 3.2 and 3.4 - Verify the obtained circuits by simulation



	amax of	amin dB	f0 Hz	f_s Hz
3.1	0.5	30	1000	2330
3.2	0.5	20	1000	2000
3.3	1.0	35	1000	3500
3.4	0.5	20	1000	1725

Table 1

Filter specifications.

Low-pass filter attenuation template.

3-1)

Ej 3

1) $\alpha_{max} = 0.5$, $\alpha_{min} = 30$, $f_p = 1k$, $f_s = 2330$

$\alpha \rightarrow$ max/min passband

$\alpha^2 = 10^{\alpha/10} = 10^{0.5/10} = 1.122$

$n = \frac{\log \left(\frac{10^{\alpha_{min}/10} - 1}{10^{\alpha_{max}/10} - 1} \right)}{2 \log \left(\frac{f_s}{f_p} \right)} = 5.32 \rightarrow 6$

b. (Cero en este caso en el 0.30 de $\alpha = 1$)

$H(j\omega) = \frac{1}{1 + [j\omega / (2\pi f_p)]^{2n}} = \frac{1}{1 + [j\omega / (2\pi \cdot 1000)]^{12}}$

$\theta = \frac{\pi}{2n} = \frac{\pi}{12} = 15^\circ$

$\theta_1 = \frac{\pi}{12} = 15^\circ$, $\theta_2 = \frac{\pi}{12} = 15^\circ$, $\theta_3 = \frac{\pi}{12} = 15^\circ$

$\omega_1 = \frac{1}{2000s}$, $\omega_2 = \frac{1}{2000s}$, $\omega_3 = \frac{1}{2000s}$

c. $\rightarrow H_1(s) = \frac{1}{s^2 + \frac{1}{Q_1}s + 1} = \frac{1}{s^2 + \frac{1}{0.518}s + 1}$

$\frac{1}{0.518} = \frac{1}{R_1 C_1} = \frac{1}{R_2 C_2} \Rightarrow C_1 = C_2 = 1.936$

$\omega_1^2 = \frac{1}{R_1 R_2 C_1 C_2} = 1 \Rightarrow C_1 C_2 = 0.97$

$\rightarrow H_2(s) = \frac{1}{s^2 + \frac{2}{\sqrt{2}}s + 1} = \frac{1}{s^2 + \frac{1.414}{1}s + 1} \Rightarrow C_1 C_2 = 0.707$

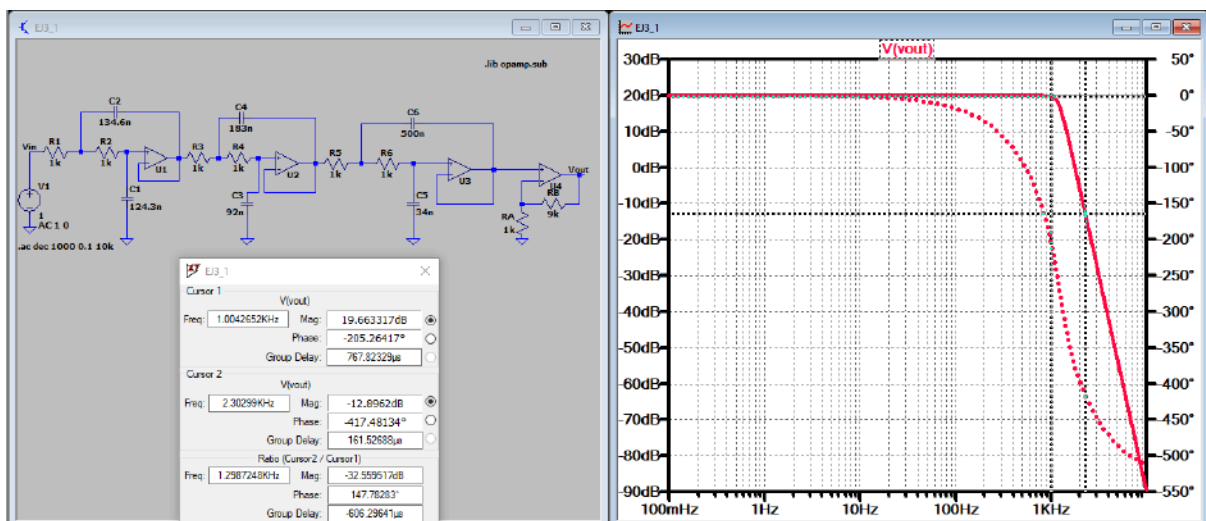
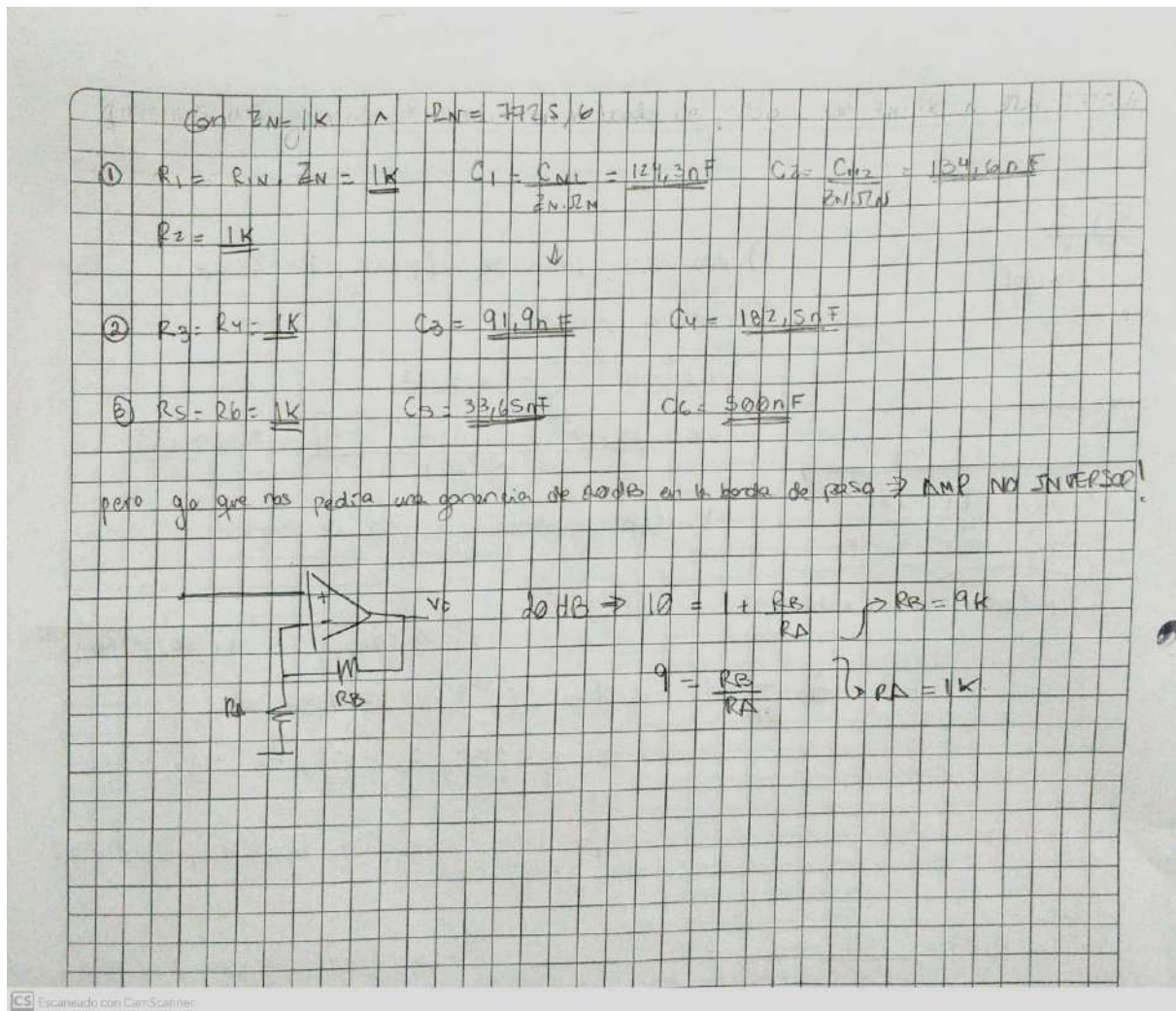
$\frac{1}{\sqrt{2}} = \frac{1}{R_3 C_3} + \frac{1}{R_4 C_4} \Rightarrow C_3 = C_4 = 1.41$

$\rightarrow H_3(s) = \frac{1}{s^2 + \frac{1}{1.932}s + 1} = \frac{1}{s^2 + \frac{1}{1.932}s + 1} \Rightarrow C_1 C_2 = 0.26$

$\frac{1}{1.932} = \frac{1}{R_5 C_5} + \frac{1}{R_6 C_6} \Rightarrow C_5 = C_6 = 3.86$

Sallen-Key $\rightarrow$ 2da orden, 0.1 Hz act. ω_0

$H(s) = \frac{1}{s^2 + \frac{1}{KQ} \left(\frac{1}{R_1 C_1} + \frac{1}{R_2 C_2} \right) s + \frac{1}{R_1 R_2 C_1 C_2}}$



Simulation of circuit 3.1 with an attenuation of approximately 0.5 dB at 1 KHz and of 32.9 dB at 2.3 KHz.

3-2)

2) $\alpha_{max} = 0,5$ $\alpha_{min} = 20$ $f_p = 2000 \text{ Hz}$ $f_s = 2 \text{ KHz}$ $W_S = 4000 \text{ rad/s}$
 $\Rightarrow W_p = 1$
 $W_{SN} = 2$

$\Rightarrow \theta^+ = 0$ $\theta^- = -180^\circ$ $\Rightarrow n = \log \left[\frac{(10^{0,1 \cdot \alpha_{max}} - 1)}{(10^{0,1 \cdot \alpha_{min}} - 1)} \right] \cdot \frac{1}{20 \log(W_S)} = 4,23 \Rightarrow 5$

$\Rightarrow \omega_c = \frac{1}{T_c} \cdot \frac{2\pi}{\omega_c} = 2254 \text{ rad/s}$
 $\Rightarrow \theta_1 = 0^\circ$ $\theta_2 = \pm 26^\circ$ $\theta_3 = \pm 72^\circ$ $\omega_c = 180^\circ / S$

$\Rightarrow Q_1 = \frac{1}{2 \cos \theta_1} = 0,5$ $Q_2 = 0,618$ $Q_3 = 1,62$

$\Rightarrow H_1(s) = \frac{1}{s^3 + 0,62s^2 + 1}$ $R_{YN} = R_{ZN} = R_{WN} = 1$
 $\cdot \frac{1}{0,62} = \frac{1}{C_1} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) \Rightarrow C_{1N} = 1,26$
 $\cdot 1 = \frac{1}{C_1 \cdot R_1 \cdot R_2} \Rightarrow C_{1N} = 0,54$

$H(s) = \frac{1}{s^3 + 0,62s^2 + 1}$ $R_{YN} = R_{ZN} = R_{WN} = 1$
 $\cdot \frac{1}{1,62} = \frac{1}{C_3} \left(\frac{1}{R_4} + \frac{1}{R_5} + \frac{1}{R_6} \right) \Rightarrow C_{3N} = 4,26$ $\cdot 1 = \frac{1}{C_3 \cdot R_4 \cdot R_5} \Rightarrow C_{3N} = 0,21$

$\Rightarrow H_2(s) = \frac{1}{s+1}$ directamente de donde obtendremos la ganancia solicitada.

$\frac{V_A - V_x}{R_1} = \frac{V_x - 0}{\frac{1}{g_m}}$ $\frac{V_B - V_x}{R_A} = \frac{V_x - 0}{R_B}$
 $\frac{V_A}{R_1} = V_x \left(g_m + \frac{1}{R_1} \right)$ $\frac{V_B}{R_A} = V_x \left(\frac{1}{R_A} + \frac{1}{R_B} \right)$
 $\frac{V_A}{R_1} = \frac{V_B}{R_A} \left(\frac{R_A}{R_1} + 1 \right)$ $V_0 = V_x \frac{R_A + R_B}{R_A R_B}$
 $\frac{V_A}{R_1} = V_0 \frac{R_A (R_B + R_1)}{R_1 (R_A + R_1)}$ $V_x = V_0 \frac{R_B}{R_A + R_B}$
 $\frac{V_0}{V_1} \frac{R_A (R_B + R_1)}{R_1 (R_A + R_1)} = \frac{R_B + R_1}{R_B (R_1 + 1)}$ $\Rightarrow \frac{V_0}{V_1} = A_v = \frac{1}{g_m R_1 + 1}$

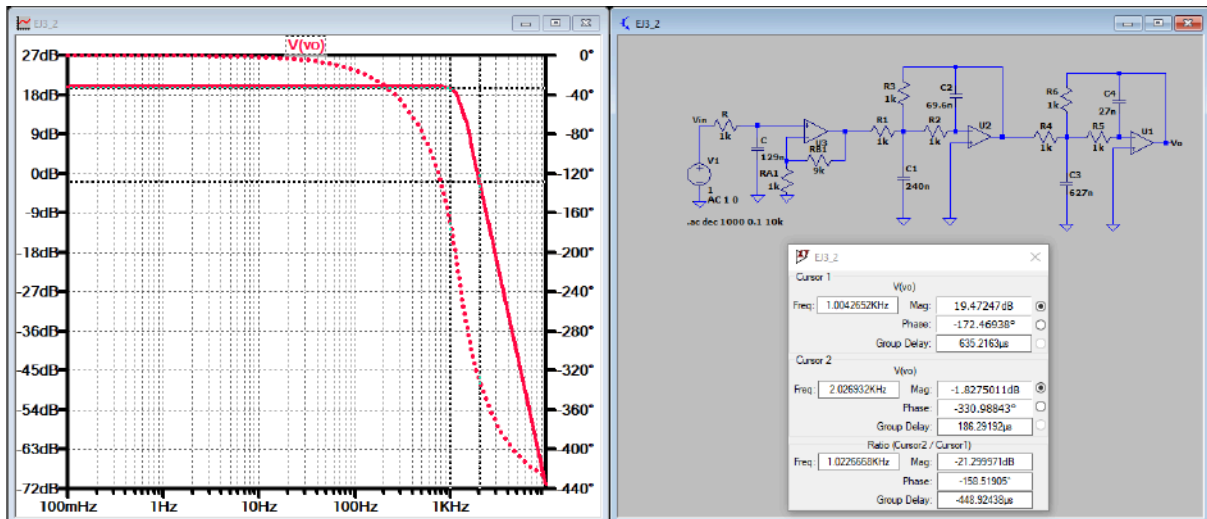
$A_v = \frac{R_B}{R_1} = \frac{R_B + R_1}{R_B}$
 $C_{N1} = 1$ $R_{N1} = 1$ $R_{A1} = 1 \text{ K}$ $R_{B1} = 9 \text{ K}$

→ Des normalizado para $H_1(5) = H_1(9)$ Com $Z_M = 1K \wedge S_{CN} = 27 \Omega$

1) $R_1 = R_2 = R_3 = \underline{1K}$ $C_1 = \frac{C_{IN}}{2 \cdot S_{CN}} = \underline{239,9 nF}$ $C_2 = \underline{69,6 nF}$ $R = \underline{728 \Omega}$

2) $R_4 = R_5 = R_6 = \underline{1K}$ $C_3 = \underline{626,2 nF}$ $C_4 = \underline{27 nF}$

6) $C = \frac{C_{IN}}{2 \cdot S_{CN}} = \underline{125,9 nF}$ $R_{IN} = R_{IN}, Z_N = \underline{1K}$



Simulation of circuit 3.2 with approx. attenuation of 0.5 dB at 1 KHz and 21.8 dB at 2 KHz.

3-3)

3) $\alpha_{max} = 1$ $\alpha_{min} = 33$ $f_p = 1K$ $f_s = 3K$ $\omega_p = 2000\pi$ $\omega_s = 7000\pi$ $\frac{\omega_p}{\omega_s} = 1$ $\frac{\omega_{SN}}{\omega_S} = 3,5$

$E^2 = 10^{-1} = 0,259$ $-n = \frac{\log\left(\frac{10^{\alpha_{min}}}{10^{\alpha_{max}}}\right)}{2 \log\left(\frac{\omega_s}{\omega_p}\right)} = 3,76 \Rightarrow n = 4$

$\omega_B = E^{\frac{1}{n}} \cdot \omega_p = 7439 \text{ rad/s}$

$\theta_1 = \pm 22,5^\circ$ $\theta_2 = \pm 67,5^\circ$ $\phi_1 = \frac{1}{2 \cos \theta_1} = 0,54$ $\phi_2 = 1,31$

$H(s) = \frac{s^2}{s^2 + \frac{\omega_B}{Q}s + \omega_B^2} = \frac{1}{s^2 + \frac{1}{Q}s + 1}$

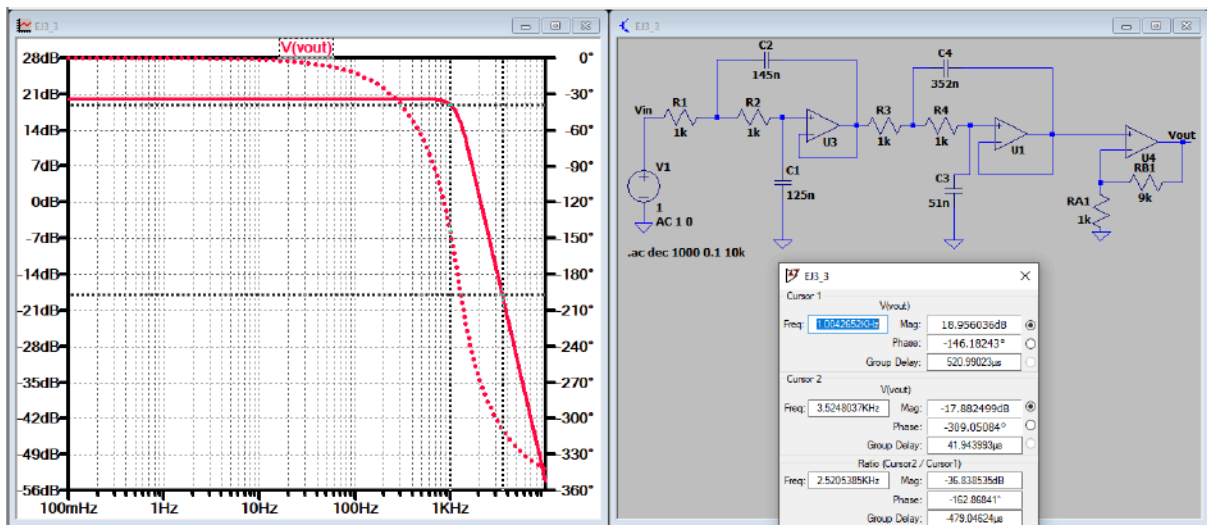
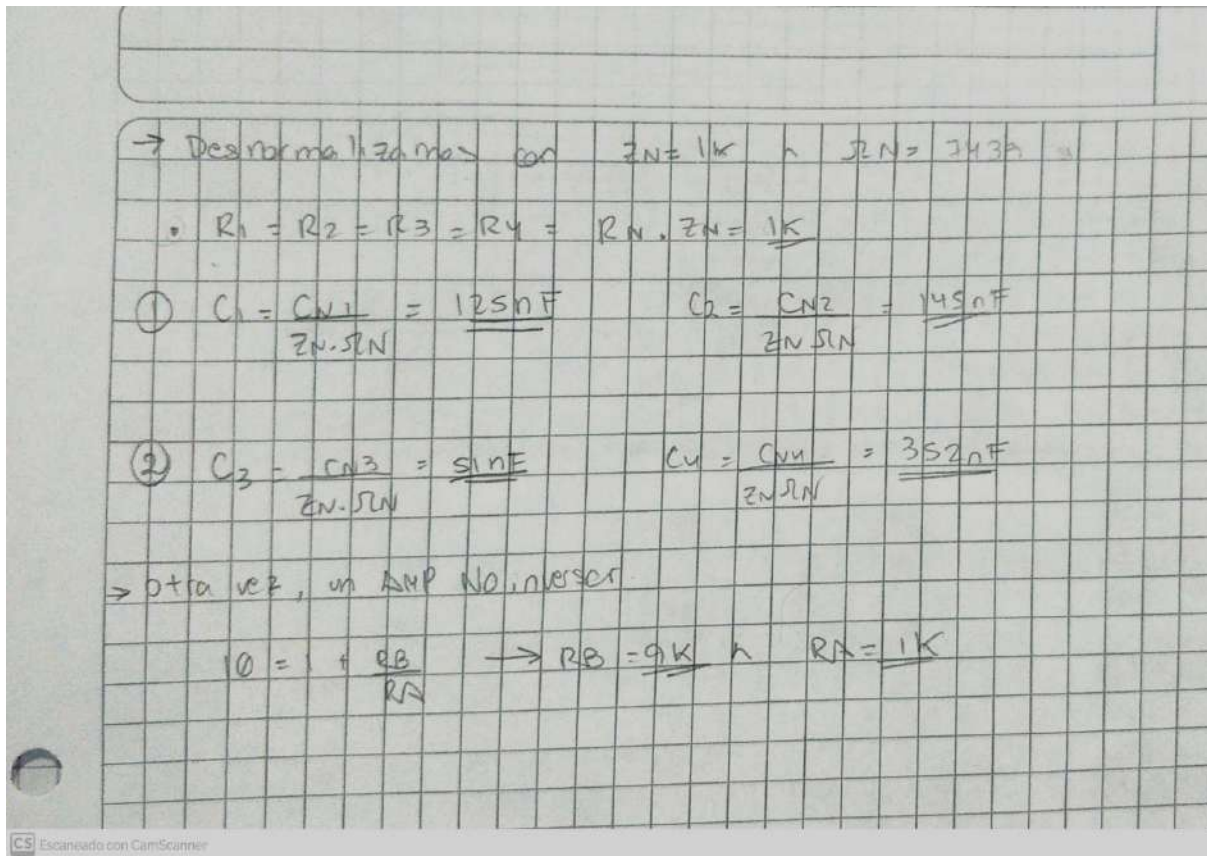
$\Rightarrow H_1(s) = \frac{1}{s^2 + \frac{1}{Q}s + 1} \rightarrow \frac{1}{0,54} = \frac{1}{R_1 C_2} + \frac{1}{R_2 C_2}$ $\text{con } R_1 = R_2 = 1$

$\frac{1}{0,54} = \frac{1}{C_2} \left(\frac{R_2 + R_1}{R_2 R_1} \right) = \frac{1}{C_2} \cdot 2 \Rightarrow C_2 = 1,08$

$\omega_B^2 = \frac{1}{R_1 R_2 C_2^2} = 1 \Rightarrow C_2 = 0,93$

$\Rightarrow H_2(s) = \frac{1}{s^2 + \frac{1}{1,31}s + 1} \rightarrow \frac{1}{1,31} = \frac{1}{R_3 C_4} + \frac{1}{R_4 C_4}$ $\text{con } R_3 = R_4 = 1$

$C_4 = 2,62$ $C_3 = 0,38$



Simulation of the circuit 3.3 with approx. attenuation of 1 dB at 1 KHz and of 37.8 dB at 3.5 KHz.

Imagen 7

3-4)

4) $d_{max} = 0.5 \text{ dB}$ $d_{min} = 20 \text{ dB}$ $f_p = 1 \text{ kHz}$ $f_s = 1723 \text{ Hz}$ $W_{PN} = 1$
 $W_P = 2000\pi$ $W_S = 3450\pi$ $W_{SN} = 1723$

$\rightarrow e^x = 10^{\frac{d_{max}-d_{min}}{20}} - 1 = 0.122 \Rightarrow n = \frac{\log[(10^{\frac{d_{max}-d_{min}}{20}} - 1)]}{2 \log(10)} = 6.14$
 $n \approx 7$

$\rightarrow J_0 = e^{\frac{1}{n}} \cdot J_P = 7302 \text{ mV}$ $180/7 = 25.71^\circ$

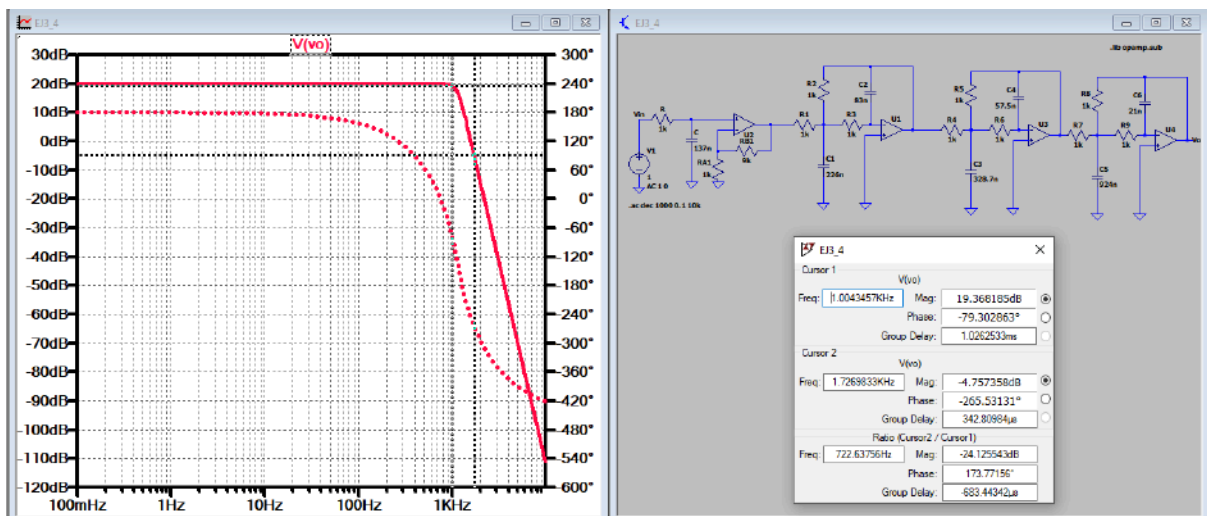
$\rightarrow \theta_1 = 25.71^\circ$ $\theta_2 = 51.43^\circ$ $\theta_3 = 77.14^\circ$ $\theta_4 = 0^\circ$

$\rightarrow Q_1 = 0.55$ $Q_2 = 0.80$ $Q_3 = 2.25$ $Q_4 = 0.5$

$H(s) = \frac{1}{(1 + sR_1C_1)(1 + sR_2C_2)(1 + sR_3C_3)(1 + sR_4C_4)}$
 $\frac{1}{Q_1} = \frac{1}{C_1(R_1 + \frac{1}{R_2} + \frac{1}{R_3})}$ con $R_4 = R_N = R_3 = 1 \Rightarrow C_1 = 1.65$
 $\frac{1}{Q_2} = 1 \Rightarrow C_2 = 0.686$
 $\frac{1}{Q_3} = \frac{1}{C_3(R_3 + \frac{1}{R_4} + \frac{1}{R_5})}$ con $R_4 = R_N = R_5 = 1 \Rightarrow C_3 = 2.4 \Rightarrow C_4 = 0.142$
 $\frac{1}{Q_4} = \frac{1}{C_4(R_4 + \frac{1}{R_5} + \frac{1}{R_6})}$ con $R_5 = R_N = R_6 = 1 \Rightarrow C_5 = 1.75 \Rightarrow C_6 = 0.145$

$\rightarrow$ Desnormalizemos: con $Z_N = 1 \text{ K} \sim J_N = 17596$
 $R_1 = R_2 = R_3 = R_4 = R_5 = R_6 = R_7 = R_8 = R_9 = R = 1 \text{ K}$ $C = \frac{1}{Z_N J_N} = 137 \text{ nF}$
 $C_1 = \frac{1.65}{Z_N J_N} = 224 \text{ nF}$ $C_2 = \frac{0.686}{Z_N J_N} = 83 \text{ nF}$ $C_3 = \frac{2.4}{Z_N J_N} = 328 \text{ nF}$ $C_4 = \frac{0.142}{Z_N J_N} = 57.5 \text{ nF}$
 $C_5 = \frac{1.75}{Z_N J_N} = 924 \text{ nF}$ $C_6 = \frac{0.145}{Z_N J_N} = 20.27 \text{ nF}$

skycolor



Simulation of the circuit 3.4 with approx. attenuation of 0.5 dB at 1 kHz and 24.8 dB at 1.75 kHz.

We see that with the Chebyshev filter we obtain a good roll-off with a lower order, in addition to better attenuation. However, the Q factors are higher and it complicates the adjustment of the circuit; on top of the ripple in the passband.

Exercise #4

A Chebyshev low-pass filter is designed to achieve an attenuation of 48 dB for frequencies greater than 9.6 kHz, with a maximum attenuation of 0.4 dB from DC to 3.2 kHz. a) Determine the order of the filter and the parameter ϵ b) Plot the magnitude response of the filter c) Determine the location of poles and zeros d) Synthesize the circuit and simulate verifying the design conditions. e) Repeat points a,b,c and d for a Butterworth filter, indicating advantages and disadvantages of both approaches.

With Sallen-Key synthesis

1) Pasabajas cheby con $att = 48 \text{ dB}$ para $f > 9.6 \text{ kHz}$ y $att_{max} = 0.4 \text{ dB}$ hasta 3.2 kHz

a- n y ϵ b- graficar $|H|$ módulo
c- polos y ceros d- sintetizar y simular
e- repetir para un Butterworth

$W_p = 6400\pi \rightarrow W_{p1} = 1 \sim W_s = 19200\pi \rightarrow W_{s1} = 3$

a- $\epsilon^2 = 10^{\frac{att_{max}}{10}} - 1 = 0.096 \rightarrow \epsilon = 0.31$ $n = \frac{\ln \left(\frac{10^{\frac{att}{10}} - 1}{10^{\frac{att_{max}}{10}} - 1} \right)}{2 \ln \left(\frac{W_s + \sqrt{W_s^2 - 1}}{W_p + \sqrt{W_p^2 - 1}} \right)} = 4.2 \rightarrow 5$

b- $|H|$

$\frac{1}{1+\epsilon^2} = 0.912$
con n IMPAR!

c- para la ubicación de polos y ceros, sigue el método del libro

$n = \frac{1}{\pi} \ln \left(\frac{1}{\epsilon^2} + \sqrt{\frac{1}{\epsilon^4} + 1} \right) = 0.378$ $\sinh(a) = 0.387$ $\cosh(a) = 1.07$

$\sigma_k = -\sinh(a) \cdot \sin\left(\frac{(2k-1)\pi}{2n}\right) \rightarrow k=1 \rightarrow -0.12$; $k=2 \rightarrow -0.31$; $k=3 \rightarrow -0.38$

$\omega_k = \cosh(a) \cdot \cos\left(\frac{(2k-1)\pi}{2n}\right) \rightarrow k=1 \rightarrow 1.02$; $k=2 \rightarrow 0.63$; $k=3 \rightarrow 0$

$P_{1,2} = -0.12 \pm 1.02j$; $P_{3,4} = -0.31 \pm 0.63j$; $P_5 = -0.38$

d- Frecuencia polo $\rightarrow \omega_k = \sqrt{\sigma_k^2 + \omega_k^2} \rightarrow \omega_{01} = 1.037$; $\omega_{02} = 0.702$; $\theta_1 = 1.106$

Factor de calidad polo $\rightarrow Q_k = \frac{\omega_k}{2\sigma_k} \rightarrow Q_{01} = 1.28$; $Q_{02} = 1.13$; $Q_2 = 0.97$

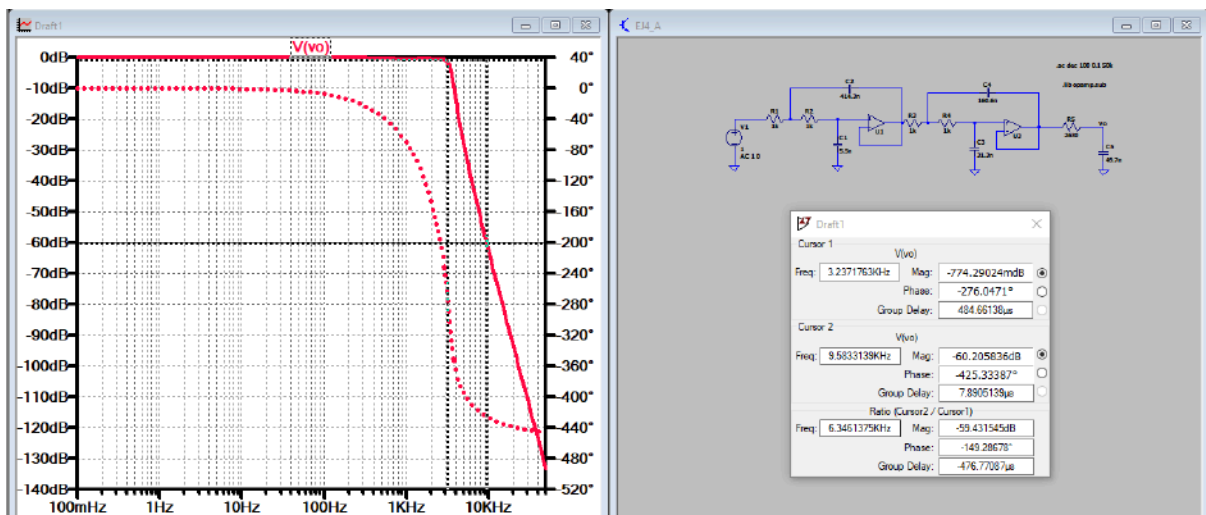
Voy a utilizar 2 estructuras sallen-key $\oplus$ una RC, y luego normalizo.

$D(s) = (s^2 + 0.78s + 1.02)(s^2 + 0.958s + 0.49)(s^2 + 0.55s + 0.49)$

$H(s) = \frac{K}{s^5 + 0.78s^4 + 1.02s^3 + 0.958s^2 + 0.55s + 0.49}$

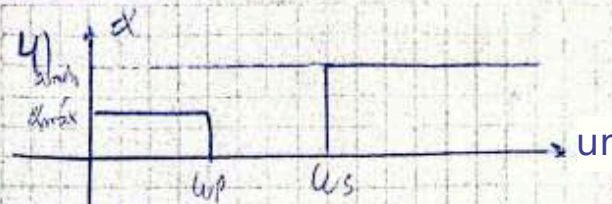
$K = \frac{1.02}{0.49} = 2.08$

$$\begin{aligned}
 & 0,24 = \frac{1}{C_2} \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad \text{con } R_1 = R_2 = 1 \Rightarrow \underline{C_{2N} = 8,33} \\
 & 1,05 = \frac{1}{R_1 R_2 C_1 C_2} \Rightarrow \underline{C_{1N} = 0,11} \\
 & \Rightarrow H(s)_2 = \frac{0,302^2}{s^2 + s \frac{0,302}{1,13} + 0,302^2} = \frac{0,49}{s^2 + s \frac{0,49}{1,13} + 0,49} \quad \text{con } R_3 = R_4 = 1 \\
 & 0,62 = \frac{1}{C_4} \left(\frac{1}{R_3} + \frac{1}{R_4} \right) \Rightarrow \underline{C_{4N} = 3,23} \\
 & 0,49 = \frac{1}{R_3 R_4 C_3 C_4} \Rightarrow \underline{C_{3N} = 0,63} \\
 & \Rightarrow H(s)_3 = \frac{1}{R_5 C_5} = \frac{0,38}{s + 0,38} \quad \text{con } R_5 = 1 \\
 & 0,38 = \frac{1}{R_5 C_5} \Rightarrow \underline{R_{5N} = 2,63} \\
 & \Rightarrow \text{Altera das normale Filter con } Z_N = 1k, f_{cN} = 20106 \\
 & R_1 = R_2 = R_3 = R_4 = \underline{1k\Omega} \quad C_1 = \frac{C_{1N}}{Z_N f_{cN}} = \underline{5,47nF} \quad C_2 = \underline{414,30F} \\
 & R_5 = \underline{2630\Omega} \quad C_3 = \underline{31,3nF} \quad C_4 = \underline{160,6nF} \quad C_5 = \underline{49,7nF}
 \end{aligned}$$



Chebyshev filter simulation with attenuation of approximately 0.4 dB at 3.2 kHz and 60 dB at 9.6 kHz.

With MFB synthesis



- Pass Band $0 \leq \omega_r \leq \omega_p$
- Transition Band $\omega_p \leq \omega_r \leq \omega_s$
- Stop Band $\omega_s \leq \omega_r$

$$\Omega_N = 32002\pi$$

$$\alpha_{\max} = 0.4 \text{ dB}, \omega_p = 2\pi \cdot 3200, \omega_s = 2\pi \cdot 32000$$

$$\alpha_{\min} = 48 \text{ dB}, \omega_s = 2\pi \cdot 32000, \omega_p = 2\pi \cdot 3200$$

$$N_n = WPLS \cdot \omega_p = 3.200$$

$$\Omega_{CN} = \frac{\omega_p - \omega_s}{\omega_p} = 1.000$$

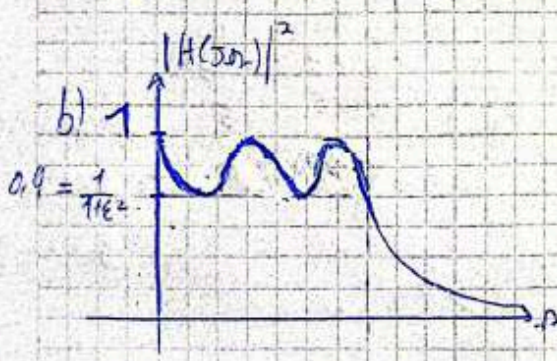
$$\Omega_{SN} = \frac{\omega_s}{\omega_p} = 10.000$$

2)

$$m \approx \frac{\ln \sqrt{4 \cdot (10^{\frac{\alpha_{\max} - \alpha_{\min}}{20}} - 1) / (10^{\frac{\alpha_{\min} - \alpha_{\max}}{20}} - 1)}}{\ln (\Omega_{SN} + \sqrt{\Omega_{SN}^2 - 1})} \approx 4.2 \approx 5$$

→ sort de pôles du filtre

$$\epsilon^2 = 10^{\frac{\alpha_{\max} - \alpha_{\min}}{20}} - 1 \Rightarrow \epsilon = 0.31$$



$$m=5 \rightarrow 5 \text{ poles} \rightarrow |H(0)| = 1$$

$$m=4 \rightarrow 4 \text{ poles}$$

$$\alpha = \frac{1}{m} \ln \left(\frac{1}{\epsilon} + \sqrt{\frac{1}{\epsilon^2} + 1} \right) = 0.378$$

$$\sigma_k = -\sin \left(\frac{(2k-1)\pi}{2m} \right) \cdot \sinh(\alpha) \rightarrow K=1 \Rightarrow \sigma_{1,2} = -0.12$$

$$\rightarrow K=2 \Rightarrow \sigma_{3,4} = -0.3131$$

$$\rightarrow K=3 \Rightarrow \sigma_5 = -0.387$$

$$\Omega_k = \cos \left(\frac{(2k-1)\pi}{2m} \right) \cdot \cosh(\alpha) \rightarrow K=1 \Rightarrow \Omega_{1,2} = \pm 1.02$$

$$\rightarrow K=2 \Rightarrow \Omega_{3,4} = \pm 0.63$$

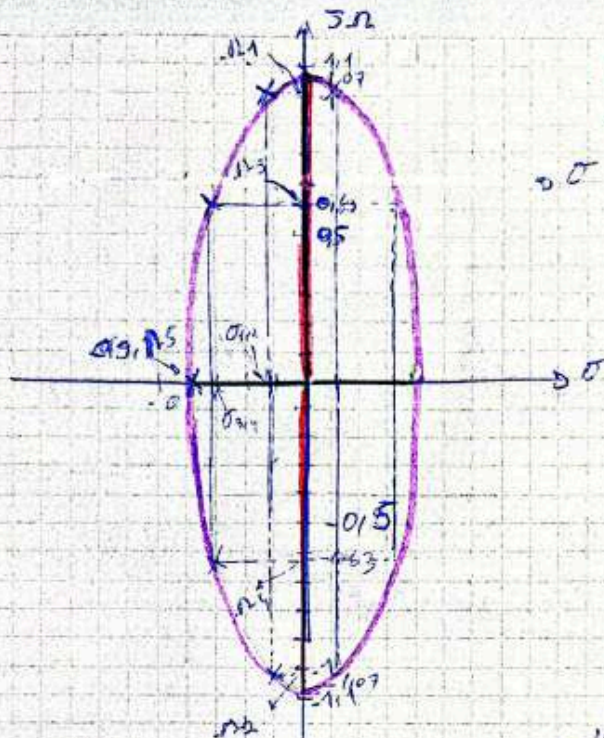
$$\rightarrow K=3 \Rightarrow \Omega_5 = 0$$

$$\sinh(\alpha) = \frac{e^\alpha - e^{-\alpha}}{2}$$

$$\cosh(\alpha) = \frac{e^\alpha + e^{-\alpha}}{2}$$

$$\sinh(\alpha)^{-1} = \ln(2 + \sqrt{4 + 1})$$

$$\cosh(\alpha)^{-1} = \ln(2 + \sqrt{2 + 1}), \alpha \geq 1$$



$$\bullet \cosh(2) = 1.07$$

$$\bullet \sinh(2) = 0.357$$

$\rightarrow \sigma$

• Factor q de cada polo

$$q_k = \frac{n_{0k}}{o_{0k} \cdot 2}$$

$$4.28 = 0.1, 2$$

$$1.13 = 0.314$$

$$0.5 = 0.5$$

• Distancias de cada polo al origen $\Rightarrow$

$$n_{0k} = \sqrt{\sigma_k^2 + \omega_k^2}$$

$$\bullet n_{01} = 1.02$$

$$\bullet n_{02} = 0.707$$

• Ángulo de cada par de polos $\sigma_k = \arctg\left(\frac{\omega_k}{\sigma_k}\right)$

$$\bullet \theta_{1,2} = \pm 83.3^\circ$$

$$\bullet \theta_{3,4} = \pm 63.4^\circ$$

$$\bullet \theta_5 = 0^\circ$$

$$\bullet \theta_{01} = 0.707$$

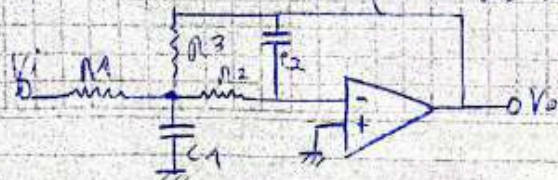
$$\bullet \theta_{02} = 0.707$$

d) Síntesis por Múltiple Feedback:

• Plantas Transmisoras de FPB de Orden 2: $H(s) = \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$

• Plantas Transmisoras de Filtro OPAMP NFM Orden 2, 2:

$$H(s) = \frac{-1 / R_1 R_2 C_1 C_2}{s^2 + s \frac{R_1}{C_1} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) + \frac{1}{R_2 R_3 C_1 C_2}}$$



Planteo transpuesta de un RC. TPB orden m=1

$$H(s) = \frac{W_0}{s + W_0}, W_0 = \frac{1}{R \cdot C}$$

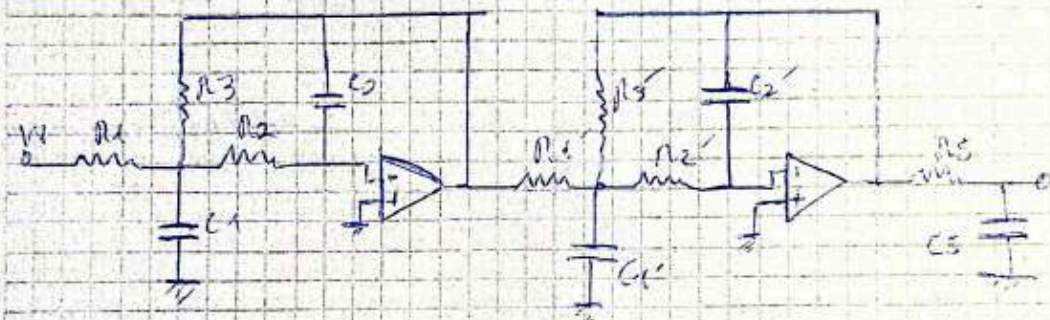
Por inspección entre ambos transfuncionales queda:

$$\Delta K^2 = \frac{1}{R_1 \cdot R_2 \cdot C_1 \cdot C_2}$$

$$\frac{\Delta K}{K_0 K} = \frac{1}{C_1} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$$

$$\Delta K = \frac{1}{R \cdot C}$$

Circuitos:



De (1) $\Delta_{1,2,3} = 1 \text{ m}$, $Z_N = \Delta K$ - por criterio de diseño

$$\Rightarrow \text{Por de } \Delta_{1,2,3} (C_1, R_1, R_2) \Rightarrow \frac{\Delta_{1,2}}{K_0 \Delta_{1,2}} = \frac{1}{C_1} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$$

$$\Rightarrow \frac{1,027}{1,125} = \frac{1}{C_1} \left(\frac{1}{1} + \frac{1}{1} + \frac{1}{1} \right) \Rightarrow C_1 = 1,5 \text{ F}$$

$$\text{y aplicando } \Delta_{1,2}^2 = \frac{1}{R_1 R_2 R_3 C_1 C_2} \Rightarrow C_2 = 0,0758 \text{ F}$$

Desnormalizando: (con $R_N = 3200 \cdot 2\pi$)

$$C_1 = \frac{C_1}{Z_N R_N} = 62,17 \text{ mF} \quad C_2 = \frac{C_2}{Z_N R_N} = 8,177 \text{ mF} \Rightarrow C_2$$

$$\Delta_{1,2,3} = \Delta_{1,2,3} N \quad Z_N = 1 \text{ K} \Rightarrow \Delta_{1,2,3}$$

De ② $\boxed{R_{123N} = 1\Omega}$ y $\boxed{Z_N = 1K}$ por criterios de diseño.

$$\Rightarrow \text{Por ende, polos } P_{3,4} = (\sigma_{3,4}, \Omega_{3,4}) \Rightarrow \Omega_{3,4} = \frac{1}{C_{1N}} \left(\frac{1}{R_{1N}} + \frac{1}{R_{2N}} + \frac{1}{R_{3N}} \right)$$

$$\Rightarrow \frac{0.702}{4.43} = \frac{1}{C_{1N}} \cdot \left(\frac{1}{2} + \frac{1}{1} + \frac{1}{1} \right) \Rightarrow \boxed{C_{1N} = 4.837}$$

$$\text{y aplicando } \Omega_{3,4}^2 = \frac{1}{R_{1N} R_{2N} C_{1N} C_{2N}} \Rightarrow \boxed{C_{2N} = 0.42F}$$

Desnormalizando con $\Omega_N = 3200 \cdot 2\pi$

$$C_1 = \frac{C_{1N}}{Z_N \cdot \Omega_N} = \boxed{240.22 \text{ mF}} \quad , \quad C_2 = \frac{C_{2N}}{Z_N \cdot \Omega_N} = \boxed{20.9 \text{ mF}} \rightarrow C_2'$$

$$R_{123}' = R_N \cdot Z_N = \boxed{1K\Omega = R_{123}'}$$

De ③ $\boxed{R_N = 1\Omega}$ y $\boxed{Z_N = 1K}$ por criterios de diseño.

$$\Rightarrow \text{Por ende 1 Polo } P_5 = (\sigma_5, \Omega_5) \Rightarrow \Omega_5 = \frac{1}{R_N \cdot C_N}$$

$$\Rightarrow 0.38 = \frac{1}{1\Omega \cdot C_N} \Rightarrow \boxed{C_N = 2.631F}$$

Desnormalizando con $\Omega_N = 3200 \cdot 2\pi$

$$C_5 = \frac{C_N}{Z_N \cdot \Omega_N} = \boxed{430.8 \text{ mF}} \quad , \quad R_5 = R_N \cdot Z_N = \boxed{1K = R_5}$$

Simulador Netmargui:

Ampliación > 48 dB con Simul = 60 dB y obtenemos más ruido de los pñidos en los transitorios de stop, lo cual es buena.

Ampliación > 48 dB con Simul = 60 dB y obtenemos ^{aprox} más ruido en los transitorios de stop, lo cual es malo por el ruido de fondo de 0.15 dB de diferencia no es tan grave.

NOTA:

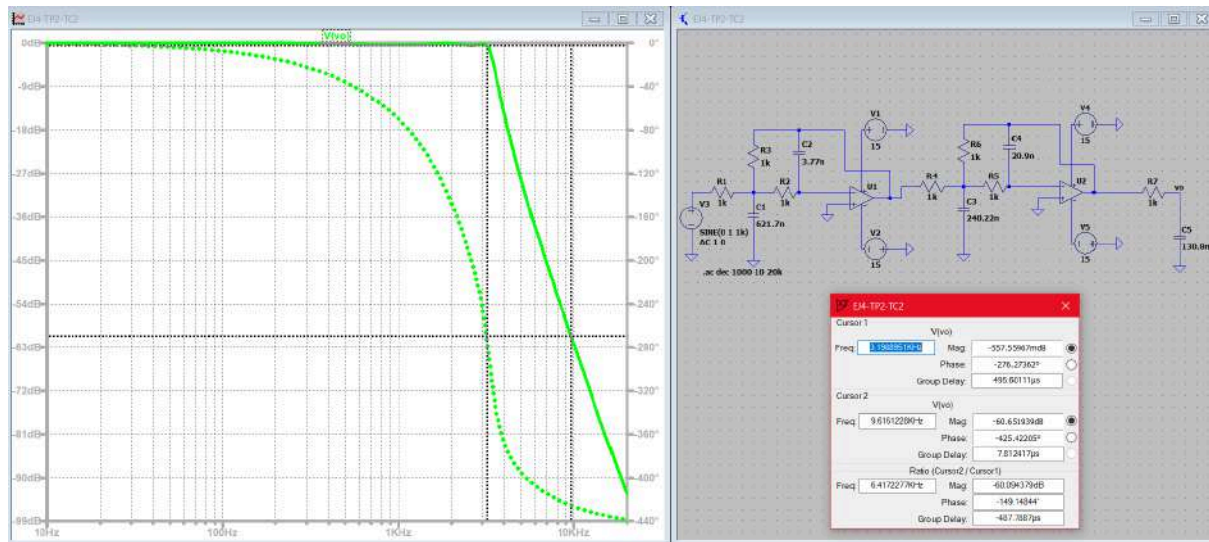


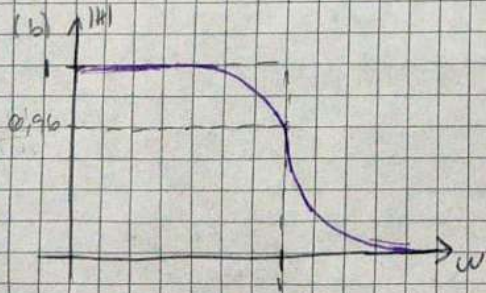
Image 9.2

Chebyshev filter simulation with attenuation approx. 0.57 dB at 3.2 KHz and 60 dB at 9.6 KHz.

Conclusion: Using Sallen-Key or Multiple Feedback synthesis, we obtain very similar results. This indicates that the obtained response mainly depends on the approximation used, which in this case was Chebyshev. It has a good roll-off and a good attenuation for a high order like $n = 5$.

Repetimos por BW.

(a) $\cdot E^2 = 0,096 \rightarrow \underline{\underline{\rho = 0,311}}$ $\cdot n = \frac{\log[(10^{0,1 \text{ dB}} - 1) / (10^{0,1 \text{ dB}} - 1)]}{2 \cdot \log(4 \omega_s)} = 6,9 \rightarrow \underline{\underline{7}}$



con $\frac{1}{\sqrt{1+E^2}} = 0,96$ $\omega_B = \frac{1}{8} \cdot \omega_s$ $\omega_P = 23757$

$D(s) = (s+1)(s^2 + \frac{1}{0,55}s + 1)(s^2 + \frac{1}{0,8}s + 1)(s^2 + \frac{1}{2,25}s + 1)$

(c) $\cdot \theta_1 = -25,7^\circ$ $\cdot \theta_2 = -5,4^\circ$ $\cdot \theta_3 = 77,14^\circ$ $\cdot \theta_4 = 0^\circ$ $\cdot \theta_5 = 0^\circ$
 $\cdot Q_1 = 0,55$ $\cdot Q_2 = 0,8$ $\cdot Q_3 = 2,25$ $\cdot Q_4 = 0,5$

$\left. \begin{array}{l} P_{1-2} = (-0,9 \pm 0,42i) \cdot \omega_B \\ P_{3-4} = (-0,635 \pm 0,78i) \omega_B \\ P_{5-6} = (-0,22 \pm 0,97i) \omega_B \end{array} \right\}$

$P_7 = (-1) \cdot \omega_P$

skycolor

IV) Igual que antes, usando una combinación de RC y de llenos-ley

$\rightarrow H_1(s) = \frac{1}{s^2 + \frac{1}{0,55}s + 1}$ con $R_N = R_{2N} = 1$

$\cdot \frac{1}{0,55} = \frac{1}{C_N \left(\frac{1}{R_1} + \frac{1}{R_2} \right)} \Rightarrow \underline{\underline{C_N = 1,1}}$ $\cdot 1 = \frac{1}{C_N R_N R_2} \Rightarrow \underline{\underline{C_N = 0,909}}$

$\rightarrow H_2(s) = \frac{1}{s^2 + \frac{1}{0,8}s + 1}$ con $R_N = R_{4N} = 1$

$\cdot \frac{1}{0,8} = \frac{1}{C_N \left(\frac{1}{R_3} + \frac{1}{R_4} \right)} \Rightarrow \underline{\underline{C_N = 1,6}}$ $\cdot 1 = \frac{1}{C_N R_N R_2} \Rightarrow \underline{\underline{0,625 = C_N}}$

$\rightarrow H_3(s) = \frac{1}{s^2 + \frac{1}{2,25}s + 1}$ con $R_N = R_{6N} = 1$

$\cdot \frac{1}{2,25} = \frac{1}{C_N \left(\frac{1}{R_5} + \frac{1}{R_6} \right)} \Rightarrow \underline{\underline{C_N = 4,5}}$ $\cdot 1 = \frac{1}{C_N R_N R_6} \Rightarrow \underline{\underline{0,22 = C_N}}$

$\rightarrow H_4(s) = \frac{1}{s-1} = \frac{1}{s-1}$ con $R_N = 1$ $C_N = 1$

$\rightarrow$ Ahora desnormalizamos con $Z_N = 1K$ $\omega_N = \omega_B = 23757$

$\cdot R_1 = R_2 = R_3 = R_4 = R_5 = R_6 = R_7 = \underline{\underline{1K\Omega}}$

$\cdot C_1 = \frac{C_N}{Z_N \omega_N} = \underline{\underline{38,26nF}}$ $\cdot C_2 = \underline{\underline{46,3nF}}$ $\cdot C_3 = \underline{\underline{26,1nF}}$ $\cdot C_4 = \underline{\underline{67,3nF}}$

$\cdot C_5 = \underline{\underline{9,26nF}}$ $\cdot C_6 = \underline{\underline{189,4nF}}$ $\cdot C_7 = \underline{\underline{42,1nF}}$

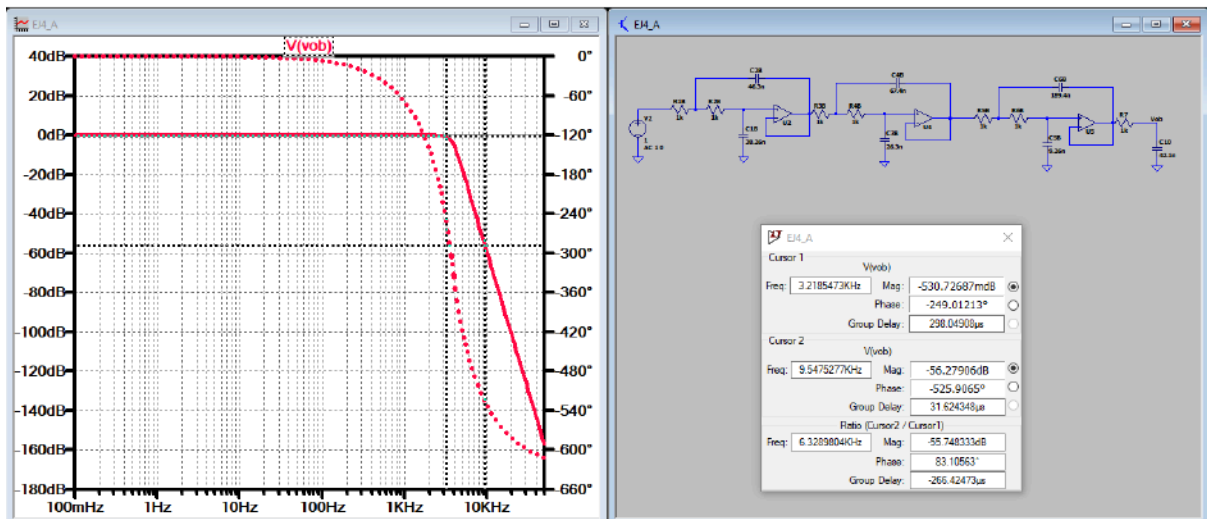


Image 10

Butterworth filter simulation with approx. attenuation of 0.4 dB at 3.2 KHz and 56 dB at 9.6 KHz.

Exercise #5

From the template in the image, knowing that: $\alpha_{\min} = 35 \text{ dB}$ $\alpha_{\max} = 1 \text{ dB}$ $f_s = 1000 \text{ Hz}$ $f_p = 3500 \text{ Hz}$, obtain poles and zeros for maximum flatness in the passband and compare with exercise 3.3.

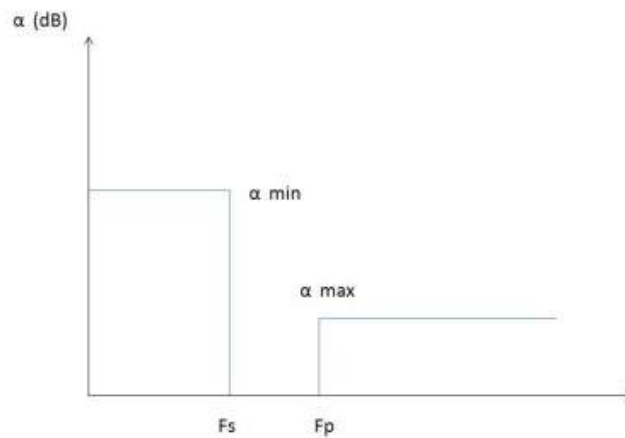


Image 11

Template of the requested filter.

e) s

a) polos y ceros para max. planicidad en la banda de paso

b) comparar con 3.3

→ Normalizamos frecuencias! $\Rightarrow \omega_p = 1$
 $\omega_s = \frac{\omega_s}{\omega_p} = \frac{2}{1}$

$\Rightarrow 8^2 - 10^{-1} \Rightarrow \epsilon = 0,509$

$\Rightarrow n = \log_{10} \left[\frac{10^{\frac{\alpha_{\min}}{20}} - 1}{10^{\frac{\alpha_{\max}}{20}} - 1} \right] = 3,76 \Rightarrow 4$

$\Rightarrow \omega_0 = 2 \quad \omega_p = 26635 \text{ rad/s}$

$\Rightarrow \theta = 22,5^\circ, Q_1 = 0,511; Q_2 = 1,30701$

- Pero como que la plantilla es un pasabajas, así que el análisis del pasabajas antes visto hay que adaptarlo

$\Rightarrow H(z)_{LP} = \frac{1}{\left(z^2 + \frac{1}{0,511}z + 1 \right) \left(z^2 + \frac{1}{1,307}z + 1 \right)}$

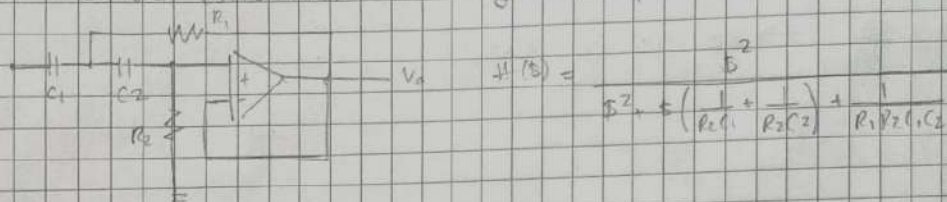
siglo 5)

$$H_{HP}(s) = H_{LP}(s) \Big|_{\substack{s = \frac{1}{s}}} = \frac{1}{\left(\frac{1}{s}\right)^2 + \frac{1}{0.541} + 1} = \frac{1}{\left(\frac{1}{s}\right)^2 + \frac{1}{s} + 1}$$

$$= \frac{1}{\left(\frac{1}{s}\right)^2 \left(1 + \frac{1}{0.541}s + s^2\right)} = \frac{1}{\left(\frac{1}{s}\right)^2 \left(1 + \frac{1}{1.31}s + s^2\right)}$$

$$= \frac{s^2}{1 + s + \frac{1}{0.541}s^2} = \frac{s^2}{1 + s + 1.31s^2}$$

Utilizo una estructura Sallen-Key HP para poder comparar con el 3.3



(1) H_{HP1} con $C_1 = C_2 = 1$

$$\frac{1}{0.541} = \frac{1}{R_2} \left(\frac{1}{C_1} + \frac{1}{C_2} \right) \Rightarrow R_2 = 1.82 \quad \frac{1}{R_1 R_2 C_1 C_2} = 1 \Rightarrow R_1 = 0.92$$

(2) H_{HP2} con $C_3 = C_4 = 1$

$$\frac{1}{1.31} = \frac{1}{R_4} \left(\frac{1}{C_3} + \frac{1}{C_4} \right) \Rightarrow R_4 = 2.62 \quad \frac{1}{R_3 R_4 C_3 C_4} = 1 \Rightarrow R_3 = 0.58$$

→ Desnormalizo con $Z_N = 1k \wedge f_N = 2000$

$$C_1 = C_2 = C_3 = C_4 = 20nF \quad R_1 = 920\Omega \quad R_2 = 182\Omega \quad R_3 = 580\Omega \quad R_4 = 2620\Omega$$

→ Analizamos como quedaron los polos!

$$H_{HP}(s) = \frac{s^4}{(s^2 + \frac{1}{0.541}s + 1)(s^2 + \frac{1}{1.31}s + 1)} \Rightarrow \text{como antes tenia 4 polos (los de los 2 filtros de 2o) se que ahora voy a tener 4 ceros en el origen}$$

$$p_{1,2} = (-0.92 \pm 0.38j) \quad p_{3,4} = (-0.38 \pm 0.92j)$$

Since none of the θ values nor the corresponding Q values change, we observe in image 12 that the pole diagram is identical to that of exercise 3.3, with the slight difference that now we have 4 zeros at the origin. This is because the pole reciprocity for the Butterworth response is direct: the poles of a low-pass filter of order "n" become the poles of a high-pass filter, and "n" zeros are added at the origin.

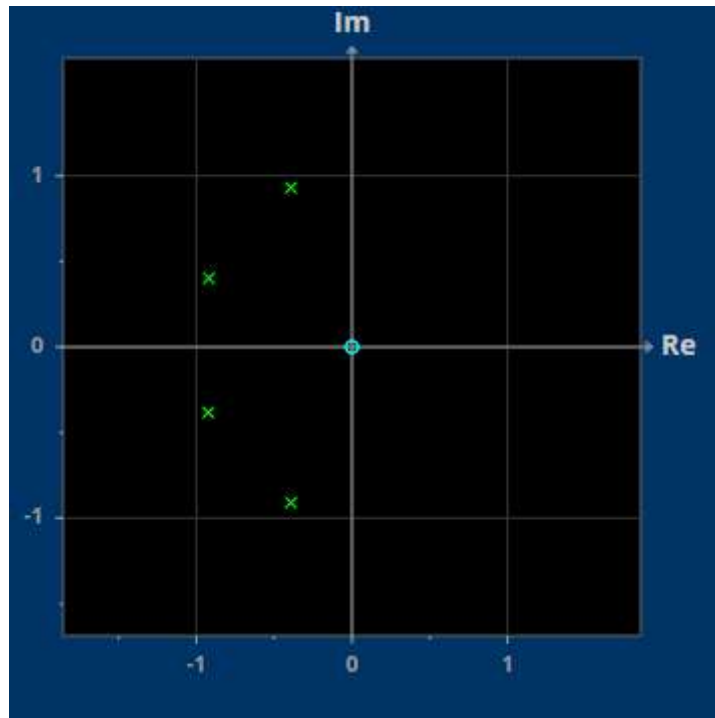


Image 12

Poles and zeros of the filter according to the sheet.

Exercise #6

Given the image template, a. Determine a filter that satisfies the template with the smallest possible order b. Compare the results with the poles of exercise 4.

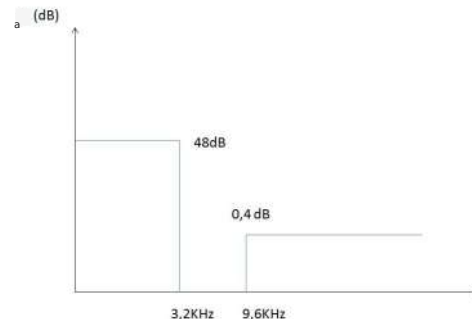
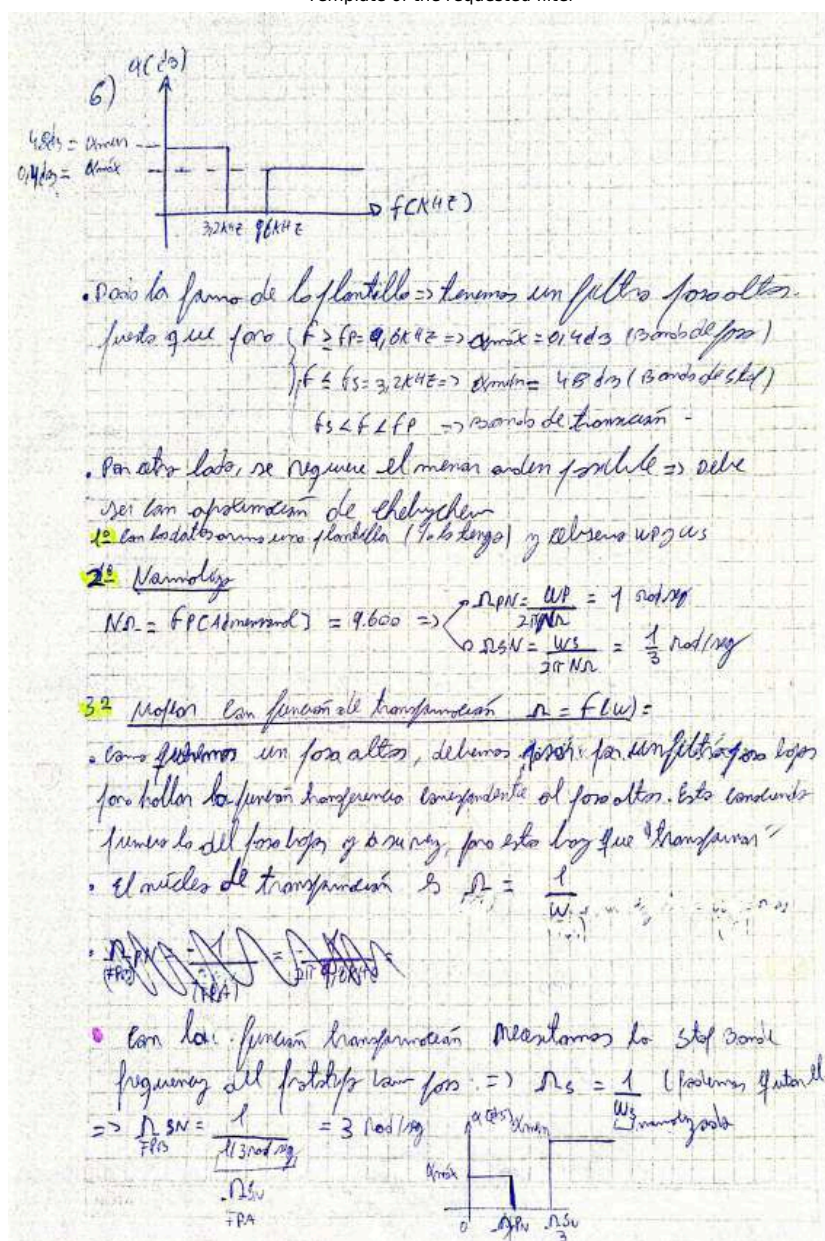


Image 13

Template of the requested filter



4.2. Buscan el orden "m" según Aproximación elegida.

• Elip Chebyshev $\Rightarrow m = \frac{\ln \sqrt{4(10^{0.1 \cdot 40} - 1) / (10^{0.1 \cdot 40} - 1)}}{\ln(\Omega_s N + \sqrt{\Omega_s^2 - 1})}$

donde $\Omega_s N = 3$, $\Omega_{stop} = 48 \text{ dB}$ y $\Omega_{pass} = 0.4 \text{ dB}$
(FPO)

$\Rightarrow m = \frac{\ln \left(\sqrt{4(10^{0.1 \cdot 40} - 1) / (10^{0.1 \cdot 40} - 1)} \right)}{\ln(3 + \sqrt{3^2 - 1})} = \frac{71385}{117627} = 4.12 \approx 5 = m$

Quedará con $m=5$

5.2. Obtener $H(s)$ can-Pon

$H(s)_{LP} = \frac{\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$, Orden $m=2$, $H(s)_{LP} = \frac{\omega_0}{s + \omega_0}$, $m=1$

Por Chebyshev

donde $\omega_0 = \Omega_{pk} = \sqrt{\Omega_K^2 + \Omega_X^2}$ y $\theta_K = \arccos\left(\frac{\Omega_K}{\Omega_X}\right)$ y $\theta = \frac{1}{m} \ln\left(\frac{1}{\epsilon} + \sqrt{\frac{1}{\epsilon^2} - 1}\right)$

donde $\Omega_K = \cos\left(\frac{(2K-1)\pi}{2m}\right)$, $\Omega_X = \cos\left(\frac{(2K-1)\pi}{2m}\right) \cdot \cosh(\theta)$, $\epsilon = 10^{-0.1 \cdot 40} - 1$

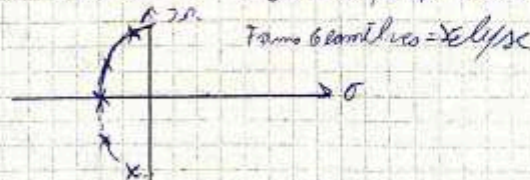
$\therefore \Omega_K = \frac{\Omega_{pk}}{\Omega_X}$, $\cosh(\theta) = \frac{e^\theta - e^{-\theta}}{2}$, $\cosh(\theta) = \frac{e^\theta + e^{-\theta}}{2}$

Del ejercicio 4)

$K=1 \Rightarrow \theta_{1,2} = -0.112$, $\Omega_{1,2} = \pm 1.02$, $\theta_{1,2} = \pm 83.3^\circ$, $\Omega_{1,2} = 1.027$, $\theta_{1,2} = 4.28$

$K=2 \Rightarrow \theta_{3,4} = -0.3131$, $\Omega_{3,4} = \pm 0.63$, $\theta_{3,4} = \pm 63.6^\circ$, $\Omega_{3,4} = 0.772$, $\theta_{3,4} = 1.13$

$K=3 \Rightarrow \theta_5 = -0.387$, $\Omega_5 = 0$, $\theta_5 = 0^\circ$, $\Omega_{0,5} = 0.38$, $\theta_{0,5} = 0.15$



6.2. Obtener $H(s)$ High-Pass

$H(s)_{HP} = H(s)_{LP} \cdot \frac{s^2}{\omega_0^2} = \frac{\omega_0^2}{\left(\frac{s}{\omega_0}\right)^2 + \frac{\omega_0}{Q} \frac{s}{\omega_0} + 1} \cdot \frac{s^2}{\omega_0^2} = \frac{s^2}{1 + \frac{\omega_0}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}}$

$H(s)_{HP} = \frac{s^2}{s^2 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{1}{\omega_0^2}}$, Orden $m=2$

$H(s)_{HP} = \frac{\omega_0^2}{s^2 + \omega_0 \frac{s}{Q} + \omega_0^2}$

$H(s)_{HP} = \frac{s^2}{1 + \frac{\omega_0}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}}$

donde $\omega_0 = \Omega_{pk}$

$H(s)_{HP} = \frac{s^2}{1 + \frac{s}{Q} + \frac{s^2}{\omega_0^2}}$

7º O usas los "pols y zeros" del tp usando los datos alternos del LP

$$H(\$/HP) = H_1(\$/HP, m=2) \cdot H_2(\$/HP, m=2) \cdot H_3(\$/HP, m=1)$$

$$H(\$/HP) = \frac{\$^2}{\$^2 + \frac{1}{0.12 \cdot 0.012} + \frac{1}{0.012^2}} \cdot \frac{1}{(0.034)^2} \cdot \frac{\$}{\frac{1}{0.5} + \$}$$

$H_1(\$/HP, m=2)$
 $H_2(\$/HP, m=2)$
 $H_3(\$/HP, m=1)$

$$H(\$/HP) = \frac{\$^2}{\$^2 + 0.2275\$ + 0.948} \cdot \frac{\$^2}{\$^2 + 1.26\$ + 2.03} \cdot \frac{\$}{2.6395 + \$}$$

$$H(\$/HP) = \frac{\$^2}{\$ - (-0.1375 + 0.965)} \cdot \frac{\$^2}{\$ - (-0.1375 - 0.965)} \cdot \frac{\$^2}{\$ - (-0.63 + 1.28)} \cdot \frac{\$}{\$ - (-0.63 - 1.28)} \cdot \frac{\$}{\$ - (-2.63)}$$

$\$1$
 $\$2$
 $\$3$
 $\$4$
 $\$5$

Pols: $\$1, \$2, \$3, \$4, \$5$

Zeros: $\$0.1, \$0.2, \$0.3, \$0.4, \$0.5 = 0$

Tenemos 5 Pols y 5 Zeros y $m=5$,

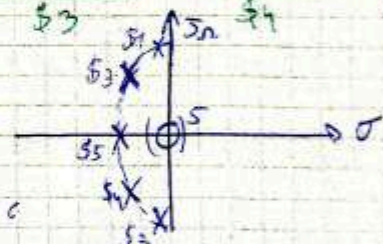
Conclusión: en Pozo alto Chelnyshu hay "m" Poles donde coincide la cantidad con el orden "m" calculado. Esto también coincide

con los "m" Poles y orden "m" para el Pozo Bajo Chelnyshu. Es decir, que tanto para Pozo alto como para Pozo bajo, la cantidad de Poles es la misma and Ad

que para ambos casos es la misma, puesto que solo depende de m y el número de condiciones de diseño).

Sin embargo, en el Pozo alto, aparecen "m" Zeros donde coincide esa cantidad con el orden "m" calculado. Esto se debe a

los en el "origen" $\Rightarrow$ tenemos un Zero de orden "5" en origen. También, notan que la ubicación de los Poles cambia en el Pozo alto respecto del Pozo Bajo, aunque la distribución geométrica sigue siendo una elipse.



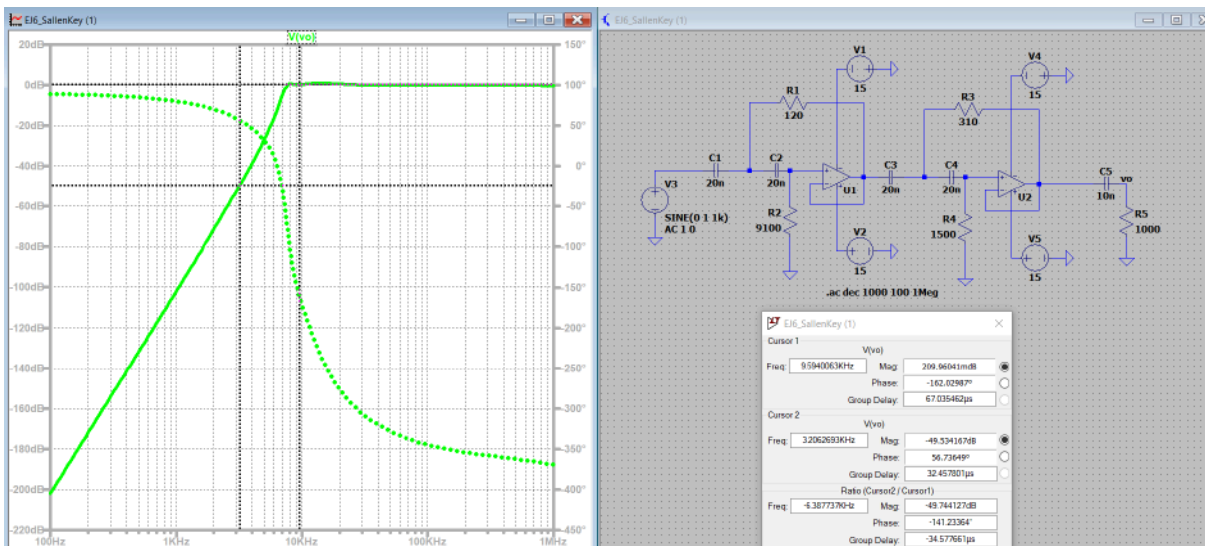
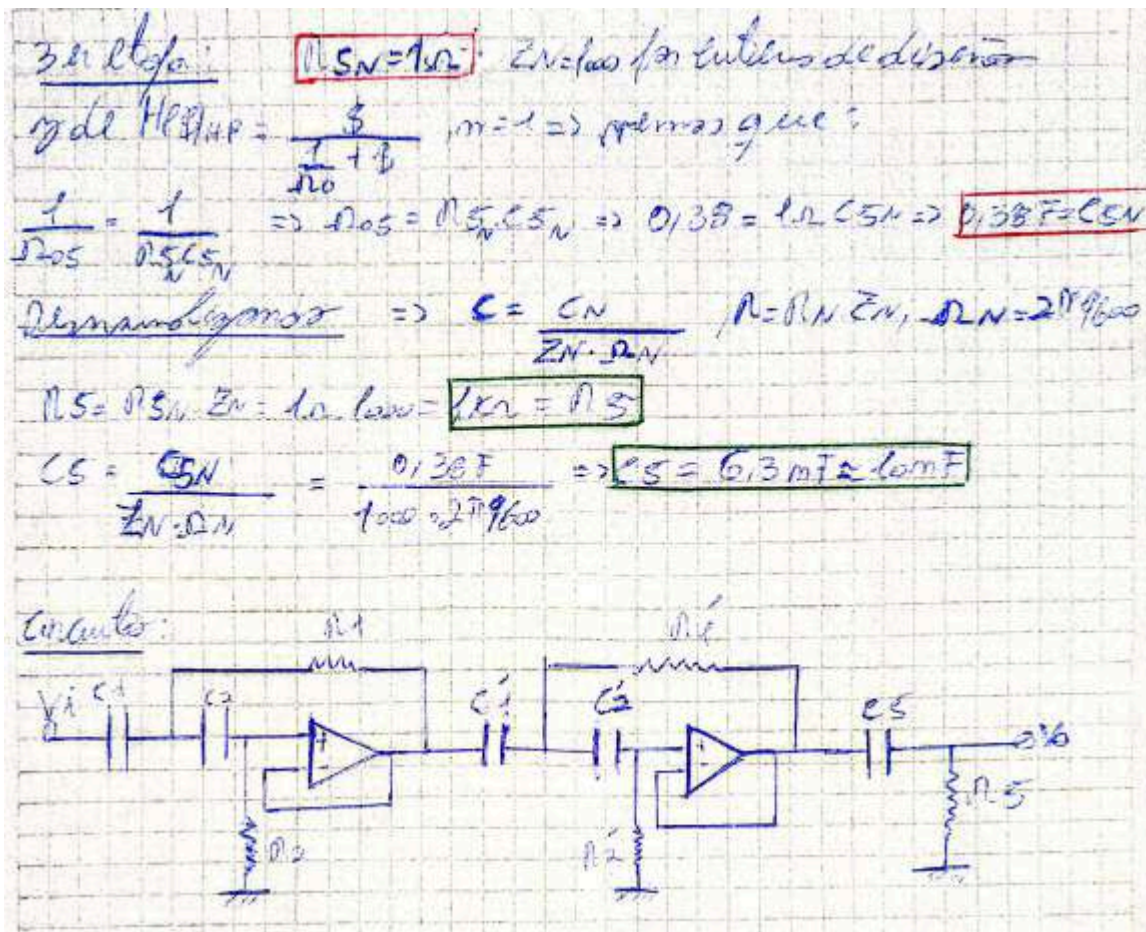


Imagen 14

Filter simulation. It is a high-pass with Chebyshev approximation to have the smallest possible "n".
Attenuation approx. of 0.4 dB at 9.6 KHz and 48 dB at 3.2 KHz.

Exercise #7

Design a low-pass filter based on an MFB structure knowing that it must provide a constant delay of 100 μ s in the passband, with a maximum delay deviation of 10% for the frequency $\omega_1=25$ krad/s and a maximum attenuation of 1 dB at the frequency $\omega_2=10$ krad/s.

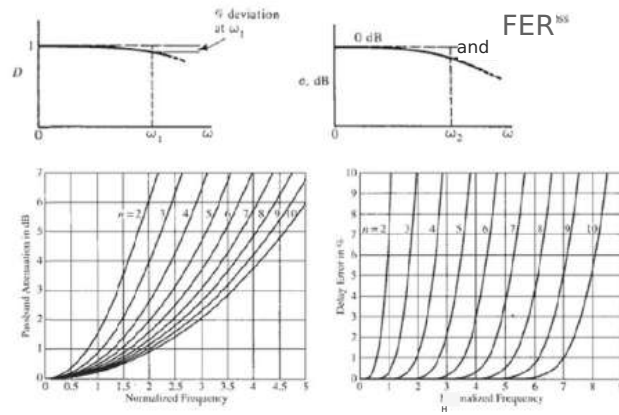


Image 15

Passband attenuation and delay error as a function of normalized frequency

7) FFB MFB con retardo cte de 100 μ s en la banda de paso con un desvío de 10% max para $\omega_1 = 25$ krad/s y 1 dB max a $\omega_2 = 10$ krad/s.

$\rightarrow$ Normalizamos con $\omega_c = 1$ $\Rightarrow \frac{\omega}{\omega_c} = \frac{1}{100 \mu s} = 10^4$ rad/s. $\rightarrow \omega_1 N = \omega_1 / \omega_c = 2.5$
 $\rightarrow \omega_2 N = \omega_2 / \omega_c = 1$

A partir del gráfico (D), vemos que con ex. nos da como $n=4$ pero si seguimos el gráfico (e), el orden se acerca más a 2. Usamos un promedio de $n=4$.

$\rightarrow$ normalizada

$\rightarrow \text{cosh}(s) = \frac{1}{s} + \frac{1}{\frac{2}{s}} + \frac{1}{\frac{5}{s} + \frac{1}{s}} = \frac{1}{s} + \frac{1}{\frac{3}{s}} + \frac{1}{\frac{5}{s} + \frac{1}{s}} = \frac{1}{s} + \frac{1}{\frac{3}{s}} + \frac{1}{\frac{5s+1}{s}} = \frac{1}{s} + \frac{s}{3} + \frac{s}{5s+1} = \frac{1}{s} + \frac{s}{3} + \frac{s}{5s+1}$

$= \frac{1}{s} + \frac{\frac{3}{s} + \frac{2.5}{s}}{\frac{3s+1}{s}} = \frac{1}{s} + \frac{3+2.5}{3s+1} = \frac{1}{s} + \frac{5.5s+3}{3s+1} = \frac{10s+10s^3+3s^2+3s^4}{10s^3+10s^2+3s+1} = \frac{\cosh(s)}{\sinh(s)}$

$\Rightarrow B_v(s) = s^4 + 10s^3 + 45s^2 + 105s + 105$

$\Rightarrow T_v(s) = \frac{B_v(0)}{B_v(s)} = \frac{105}{s^4 + 10s^3 + 45s^2 + 105s + 105}$

$T_v(s) = \frac{105}{(s^2 + 5.249s + 9.137)(s^2 + 4.749s + 11.467)}$

$\rightarrow$ Recordando que $\frac{V_o(s)}{V_i(s)} = - \frac{C_1 C_2 R_1 R_2}{s^2 + \frac{1}{C_1}(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}) + \frac{1}{C_1 C_2 R_2 R_3}}$ y que

$\frac{\omega_0}{Q} = \frac{1}{C_1}(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3})$ $\wedge$ $\omega_0^2 = \frac{1}{C_1 C_2 R_2 R_3}$

(A) Definido $R_1 = R_2 = R_3 = 1k\Omega \rightarrow R_N = 1$

$$\cdot 5,793 = \frac{1}{C_1} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) = \frac{1}{C_1} \cdot 3 \Rightarrow C_{1N} = 0,518$$

$$\cdot 9,139 = \frac{1}{C_2 \cdot C_3 \cdot R_1 \cdot R_2} = \frac{1}{C_2 \cdot 0,518} \text{ sia } \Rightarrow C_{2N} = 0,211$$

(B) $R_4 = R_5 = R_6 = 1k\Omega$

$$\cdot 4,2^{eh} = \frac{1}{C_3} \left(\frac{1}{R_4} + \frac{1}{R_5} + \frac{1}{R_6} \right) = \frac{1}{C_3} \cdot 3 \Rightarrow C_{3N} = 0,713$$

$$\cdot 11,483 = \frac{1}{C_3 \cdot C_4 \cdot R_4 \cdot R_6} = \frac{1}{C_4 \cdot 0,713} \Rightarrow C_{4N} = 0,122$$

$\rightarrow$ Desnormalizar las todas las componentes con $Z_N = 1000 \Omega$ y $A_N = 10k$

$$\cdot R_1 = R_2 = R_3 = R_4 = R_5 = R_6 = R_N \cdot Z_N = 1k\Omega$$

$$\cdot C_1 = \frac{C_{1N}}{R_N \cdot Z_N} = \frac{15h}{10k} = 15h \cdot 10^{-6} F$$

$$\cdot C_2 = \frac{C_{2N}}{R_N \cdot Z_N} = \frac{21,1n}{10k} = 21,1nF$$

$$\cdot C_3 = \frac{C_{3N}}{R_N \cdot Z_N} = \frac{71,3n}{10k} = 71,3nF$$

$$\cdot C_4 = \frac{C_{4N}}{R_N \cdot Z_N} = \frac{12,2n}{10k} = 12,2nF$$

(Esto se decidió por facilidad de cálculo, pero en la práctica sería mejor fijar los capacitores primero según los valores disponibles)

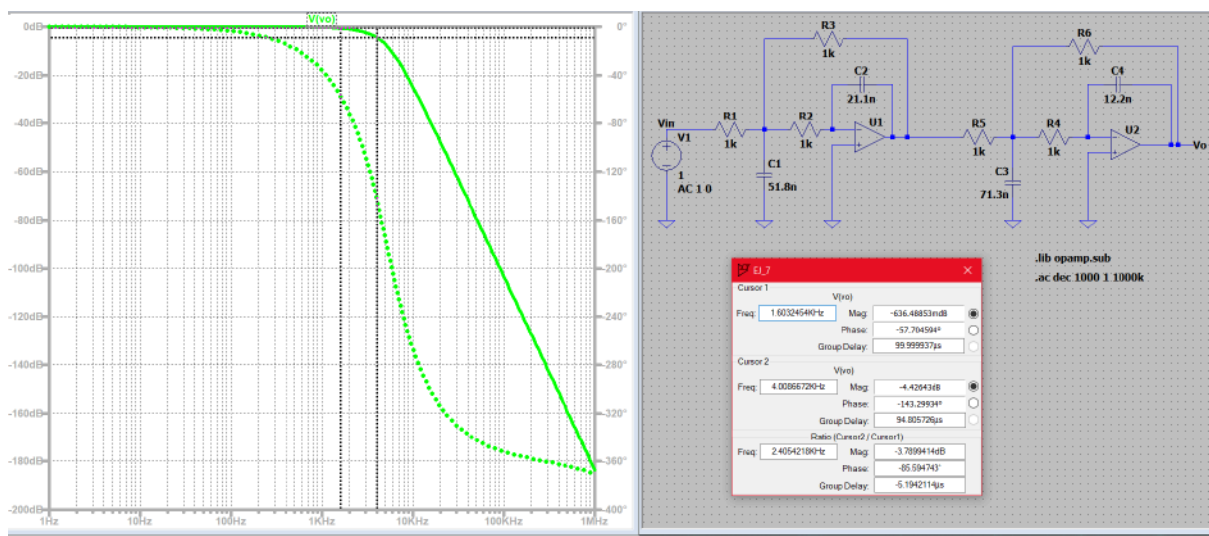


Image (a)

Filter simulation. It is a low-pass with Bessel approximation, conditions per statement. Approximate attenuation of 0.63 dB at $f_c = 1.6$ kHz or its $\omega_1 = 10$ krad/s equivalent; and 4.42 dB at $f_s = 4$ kHz or its equivalent of $\omega_2 = 25$ krad/s. Of the attenuations, on average a maximum ripple of 3.79 dB.

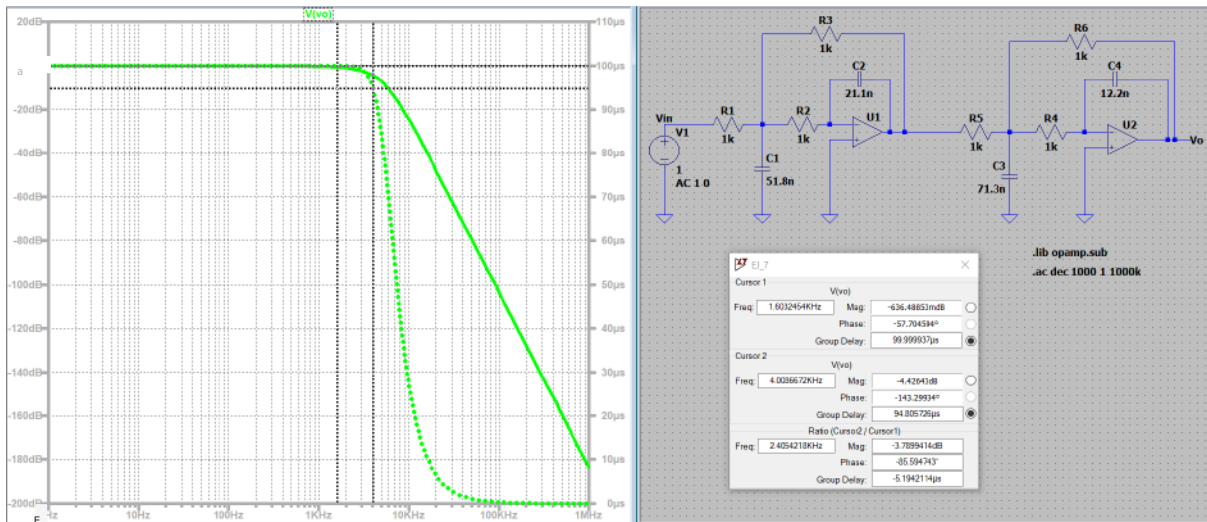


Image (b)

Filter simulation. It is a low-pass filter with Bessel approximation, conditions as stated. Group Delay of 99.997 us at $f_c = 1.6$ KHz and 94.80 us at $f_s = 4$ KHz. From the attenuations, on average a Group delay of 5.2 us

Exercise #8

A band-pass filter must be designed with the following specifications: ● Cutoff frequency lower: 1600 KHz ● Upper cutoff frequency: 2500 KHz ● Maximum ripple in the passband: 3dB ● Maximum flatness in the passband ● Maximum gain in the passband: 10 dB ● Minimum attenuation of 20 dB at frequencies of 1250 KHz and 3200 KHz

KHz. Requirements: a. Obtain the normalized transfer function of the filter b. Plot the pole-zero diagram c. Plot the transfer function (magnitude and phase) of the requested filter d. Synthesize the filter using Ackerberg Mossberg structures e. Simulate the obtained filter, verifying the

design specifications

8) $f_{ci} = 1600 \text{ KHz}$, $f_{cs} = 2500 \text{ KHz}$

Notes: $\left\{ \begin{array}{l} \text{Ripple máx en banda de paso} = 3 \text{ dB} = \alpha_{\text{máx}} = 3 \text{ dB} \\ \text{Máxima planicidad en banda de paso} \\ \text{Ganancia máxima} = \alpha_{\text{máx}} = 10 \text{ dB en banda de paso} \\ \text{Atenuación mínima} = \alpha_{\text{mín}} = 20 \text{ dB para } f_{ci} = 1250 \text{ KHz}, f_{cs} = 3200 \text{ KHz} \end{array} \right.$

10) Plantea plantilla y busca ω_0 , Q y BW

$\omega_0 = \sqrt{\omega_{ci} \cdot \omega_{cs}}$, $\omega_{ci} = 2\pi f_{ci}$, $\omega_{cs} = 2\pi f_{cs}$

$\omega_0 = \sqrt{2\pi \cdot 1600 \text{ KHz} \cdot 2\pi \cdot 2500 \text{ KHz}}$

$\omega_0 = 12,56 \text{ Mrad/s} = 12,56637 \text{ Krad/s}$

$BW = \omega_{cs} - \omega_{ci} = 2\pi \cdot 2500 \text{ KHz} - 2\pi \cdot 1600 \text{ KHz} = 1800\pi = 5,6548 \text{ Mrad/s} = BW$

$Q = \frac{\omega_0}{BW} = \frac{12,56 \text{ Mrad/s}}{5,6548 \text{ Mrad/s}} = 2,22 = Q$

12) Normalizamos la plantilla del filtro obteniendo con respecto a ω_0

$\Omega_N = \omega_0 \text{ (adimensional)} = 12,56 \text{ M}$, $\Omega_N = 12,56 \text{ Mrad/s} = \omega_0$

$\Omega_{ci} = \frac{\omega_{ci}}{\Omega_N} = \frac{2\pi \cdot 1600 \text{ K}}{12,56 \text{ M}} = 0,8 \text{ rad/s}$, $\Omega_{cs} = \frac{\omega_{cs}}{\Omega_N} = \frac{2\pi \cdot 2500 \text{ K}}{12,56 \text{ M}} = 1,25 \text{ rad/s}$

$\Omega_{s1} = \frac{\omega_{s1}}{\Omega_N} = \frac{2\pi \cdot 1250 \text{ K}}{12,56 \text{ M}} = 0,625 \text{ rad/s}$, $\Omega_{s2} = \frac{\omega_{s2}}{\Omega_N} = \frac{2\pi \cdot 3200 \text{ K}}{12,56 \text{ M}} = 1,6 \text{ rad/s}$

3º Mapas utilizando núcleos de transformación e identificar Ω -S

Núcleos de transformación $\Rightarrow \Omega = f(w) = \frac{Ue}{Bw} \left(\frac{w}{w_0} - \frac{w_0}{w} \right)$

• Vamos que $\Omega_{S1N} \cdot \Omega_{S2N} = 0,625 \cdot 1,6 = 1 \Rightarrow$ de $f(w)$ podemos hallar Ω_{SN} y este es único $\Rightarrow \Omega_{SN} = \frac{w_0}{Bw} \left(\frac{w_{e\text{ uss}}}{w_0} - \frac{w_0}{w_{s\text{ o uss}}} \right)$

$$\Rightarrow \Omega_{SN} = f(w) = \frac{12,56 \text{ M rad/seg}}{516548 \text{ M rad/seg}} \left(\frac{211,32 \text{ K rad/seg}}{12,56 \text{ M rad/seg}} - \frac{12,56 \text{ M rad/seg}}{211,32 \text{ K rad/seg}} \right)$$

e

$$\Omega_{SN} = f(w) = \frac{12,56 \text{ M rad/seg}}{516548 \text{ M rad/seg}} \left(\frac{211,32 \text{ K rad/seg}}{12,56 \text{ M rad/seg}} - \frac{12,56 \text{ M rad/seg}}{211,32 \text{ K rad/seg}} \right)$$

$$\Omega_{SN} = \pm 2,16 \Rightarrow \text{en módulo } \Omega_{SN} = 2,16$$

4º Buses "m" y la $H(z)$ _{Banda} a partir de la $H(z)$ _{Pasa}

Se que se pide Máx. Planchidad y que tenemos de un modelo por los por llegar ^{gutterman y que ripple = -3dB} _{fora banda. entonces, vale:}

$$m = \frac{\log \left[\frac{(10^{0,1} - 1)}{(10^{0,1} - 1)} \right]}{2 \log(\Omega_{SN})} \quad \text{y vemos cuanto da m...}$$

$\alpha_{\text{máx}} = 3\text{dB}$
 $\alpha_{\text{mín}} = 20\text{dB}$

$$\text{Con } \Omega_{SN} = 2,16 \text{ rad/seg} \Rightarrow m = \frac{\log \left[\frac{(10^{0,1} - 1)}{(10^{0,1} - 1)} \right]}{2 \log(2,16 \text{ rad/seg})} = 2,98 \approx 3 = m$$

$$\text{De } H(z)_{\text{Pasa}} = \frac{1}{s^2 + \frac{w_0}{Q}s + w_0^2} \quad \text{orden 2} \quad \text{y } H(z)_{\text{Banda}} = \frac{1}{s^2 + \frac{w_0}{Q}s + w_0^2} =$$

donde reemplazamos $s = \frac{1}{T} \left(\frac{z}{z_0} + \frac{z_0}{z} \right)$, $z_0 = 1$ ^{por} _{normalizar} lo que simplifica los cuentas. $\Omega_0 = \frac{w_0}{\Omega_N} = 1$

Trochogramas algebraicamente:

$$\text{De orden } m=2 \Rightarrow H(s)_{\text{Low Pass}} = \frac{1}{1 + \frac{s}{Q_0} + \frac{s^2}{Q_0^2}} = \frac{1}{1 + \frac{s}{Q_0} + \frac{s^2}{Q_0^2}}$$

dando el tener máxima planitud y Butterworth y $Q_0 = 1$ y $W_{0LP} = \Omega_{0LP} = 1$ para diseño.

$$H(s) = \frac{1}{\left[Q_0 \left(\frac{s}{1} + \frac{1}{s}\right)\right]^2 + 1 \left[Q_0 \left(\frac{s}{1} + \frac{1}{s}\right)\right] + 1^2}$$

$$H(s) = \frac{1}{Q_0^2 \left(\frac{s^2+1}{s}\right)^2 + 1 \left[Q_0 \left(\frac{s^2+1}{s}\right)\right] + 1}$$

$$H(s) = \frac{1}{\frac{Q_0^2 \cdot (s^4 + 2s^2 + 1)}{s^2} + \frac{1 \cdot Q_0 (s^2 + 1)}{s} + 1} \cdot \frac{s^2}{s^2}$$

$$H(s) = \frac{s^2}{Q_0^2 (s^4 + 2s^2 + 1) + \frac{1}{Q_0} \cdot Q_0 (s^2 + 1)s + s^2}$$

$$H(s) = \frac{s^2}{Q_0^2 s^4 + Q_0^2 2s^2 + Q_0^2 + \frac{1}{Q_0} Q_0 s^3 + \frac{1}{Q_0} Q_0 s + s^2}$$

$$H(s) = \frac{s^2}{Q_0^2 s^4 + \frac{Q_0}{Q_0} s^3 + (Q_0^2 \cdot 2 + 1) s^2 + \frac{Q_0}{Q_0} s + Q_0^2} \cdot \frac{1/Q_0^2}{1/Q_0^2}$$

$$H(s) = \frac{(1/Q_0^2) \cdot s^2}{s^4 + \frac{1}{Q_0} s^3 + (2 + \frac{1}{Q_0^2}) s^2 + \frac{1}{Q_0} s + 1}$$

Para analizar el denominador necesitamos datos para simplificar lo siguiente.

$\Rightarrow Q_B = 2,22$ y $\frac{1}{Q} = 1$ con $Q=1$ y $q_{w,mz} = 3$ ^{del modelo bar-Por} y en la tabla 6.3 de figura 2.60 del libro podemos ver esto.

Recomendando:

$$\frac{1}{Q_B} = 0,450, \left(2 + \frac{1}{Q_B^2}\right) = 2,203, \frac{1}{Q_B^2} = 0,203$$

$$H(\$) = \frac{0,203 \cdot \$^2}{\$^4 + 0,450 \cdot \$^3 + 2,203 \cdot \$^2 + 0,450 \cdot \$ + 1}$$

$$H(\$) = 0,203 \cdot \frac{\$^2}{[\$ - (-0,040 \pm 0,81765)] \cdot [\$ - (-0,134178 \pm 1,2085)]}$$

Por otro lado,

De orden $m=1$

$$H(\$)_{\text{gen}} = \frac{1}{Q_B \left(\frac{\$}{n_0} + \frac{n_0}{\$} \right) + W_{OLP}}$$

donde al tener máxima flexibilidad y buttenworth $q_{gen} \text{ máx} = 2,0$
 $\Rightarrow \epsilon = 1$ y $W_{OLP} = R_{OLP} = 4$ por diseño.

$$H(\$) = \frac{1}{q_0 \left(\frac{q}{r} + \frac{1}{r} \right) + 1} = \frac{1}{q_0 \left(\frac{\$^2 + 1}{\$} \right) + 1} \cdot \frac{\$}{\$}$$

$$H(\$) = \frac{\$}{q_0 (\$^2 + 1) + \$} = \frac{\$}{q_0 \$^2 + q_0 + \$} \cdot \frac{1/q_0}{1/q_0}$$

$$H(\$) = \frac{(1/q_0) \cdot \$}{\$^2 + 1 \cdot \$ + 1}$$

Para racionalizar el denominador mediante cancelar datos para simplificar la cuenta

$$\Rightarrow q_0 = 2,22 \text{ y no aparece } q_0$$

Racionalizando, $\frac{1}{q_0} = 0,450$

$$H(\$) = 0,450 \frac{\$}{\$^2 + 0,450 \$ + 1} = 0,450 \frac{\$}{[\$ - (-0,225 + 0,9745)]}$$

Finalmente:

$$H(\$) = H(\$) \cdot H(\$) = \frac{(1/q_0)^2 \cdot \$^2}{\$^4 + 1 \cdot \$^3 + (2 \cdot \frac{1}{q_0}) \$^2 + 1 \cdot \$ + 1} \cdot \frac{(1/q_0) \cdot \$}{\$^2 + 1 \cdot \$ + 1} = H(\$)$$

$$H(\$) = 0,203 \frac{\$^2}{[\$ - (-0,090 \pm 0,8175)] \cdot [\$ - (-0,134178 \pm 1,2085)]} \cdot 0,405 \frac{\$}{\$ - (-0,225 \pm 0,9745)}$$

b) Se presentan con simulación en Software.

d) 5º Sintetizar con estructura Achberg $H(s) = \frac{-K \cdot \frac{1}{s} \cdot \frac{1}{s+1}}{s^2 + \frac{1}{Q}s + 1}$

$$H(s) = H_1(s) \cdot H_2(s) \cdot H_3(s)$$

1er circuito:

$$H_1(s) = \frac{-(1/0) \cdot \frac{1}{s}}{s^2 + \frac{1}{Q}s + 1} = \frac{-0.450 \cdot \frac{1}{s}}{s^2 + 0.450s + 1}, K = 0.450, \frac{1}{Q} = 0.450 \Rightarrow Q = \frac{1}{0.450} = 2.22$$

Pair

entorno de silencio $\Rightarrow R_N = 1\Omega = R_N = R_A, Z_N = 1\Omega, C_N = 1F$
 y por naturaliza del circuito Achberg $\Rightarrow \left(\frac{R_A}{K}\right) = \frac{1\Omega}{0.450} = 2.22\Omega, (Q \cdot R) = 2.22 \cdot 1\Omega = 2.22\Omega$

Desnormalizando: $E = \frac{C_N}{R_N \cdot Z_N}, R = R_N \cdot Z_N, R_N = 12.56 \text{ M}\Omega/\text{seg}$

$$C = 1F = 80 \text{ mF} = C, R = 1\Omega \cdot 1 = 1\Omega, \frac{1}{K} = \left(\frac{R_A}{K}\right) \cdot Z_N = 2.22\Omega \cdot 1 = 2.22\Omega$$

$$R_A = 1\Omega \cdot 1 = 1\Omega, (Q \cdot R) = (Q \cdot R) \cdot Z_N = 2.22\Omega \cdot 1 = 2.22\Omega$$

2do circuito

$$H_2(s) = \frac{110s}{[s - (-0.090 + j0.81765)] \cdot [s - (-0.090 - j0.81765)]} = \frac{0.450 \cdot s}{s^2 + s \cdot 0.18 + 0.6765}$$

Integrando el denominador $\text{Denom}(s) = s^2 - s(-0.090 - 0.81765) - s(-0.090 + 0.81765) + (-0.090 + 0.81765)(-0.090 - 0.81765)$

$$\text{Denom}(s) = s^2 + s \cdot 0.090 + s \cdot 0.81765 + 0.090 \cdot 0.81765 + 0.0081 - 0.6689$$

$$\text{Denom}(s) = s^2 + s \cdot 0.18 + 0.6765$$

Por entorno de

silencio $\Rightarrow R_N = 1\Omega = R_N = R_A, Z_N = 1\Omega$
 y por naturaliza del circuito Achberg $\Rightarrow \left(\frac{R_A}{K}\right) = \frac{1\Omega}{0.450} = 2.22\Omega, (Q \cdot R) = 0.57 \cdot 1\Omega = 0.57\Omega$

Por inspección: $0.18 = \frac{1}{Q} \wedge 0.6765 = \omega_0^2 \Rightarrow \omega_0 = \sqrt{0.6765} = 0.822 \text{ rad/s} \Rightarrow \frac{1}{Q} = 0.57 \Rightarrow Q = 1.75$

$$0.450 = K \cdot \omega_0 \Rightarrow K = \frac{0.450}{0.822} = 0.547, \omega_0 = \frac{1}{R \cdot C} \Rightarrow C_N = \frac{1}{R \cdot \omega_0}$$

$$C_N = \frac{1}{1\Omega \cdot 0.822} = 1.22F$$

Normalizando $C = \frac{C_N}{\Omega_N Z_N}$, $R = R_N Z_N$, $\Omega_N = 12,56 \text{ Mrad/s}$

$C = \frac{1,227}{12,56 \text{ Mrad} \cdot 1} = 97 \text{ nF}$, $R = R_A = R_N Z_N = 1 \Omega \cdot 1 = 1 \Omega$
 $\left(\frac{R}{K}\right) = \left(\frac{R}{K}\right)_N \cdot Z_N = 1826 \Omega \cdot 1 = 1826 \Omega$
 $(Q \cdot R) = (Q \cdot R)_N \cdot Z_N = 4157 \Omega \cdot 1 = 4157 \Omega$

3er caso:

$H_3(s) = 0,450 \cdot \frac{\$}{\$} = 0,450 \cdot \frac{\$}{\$}$

$$\frac{[\$ - 0,134178 + 1,2085] \cdot [\$ - (-0,134178 - 1,2085)]}{\text{Denom}(s)} \cdot \frac{\$}{\$ + 0,268356 + 1,47726s}$$

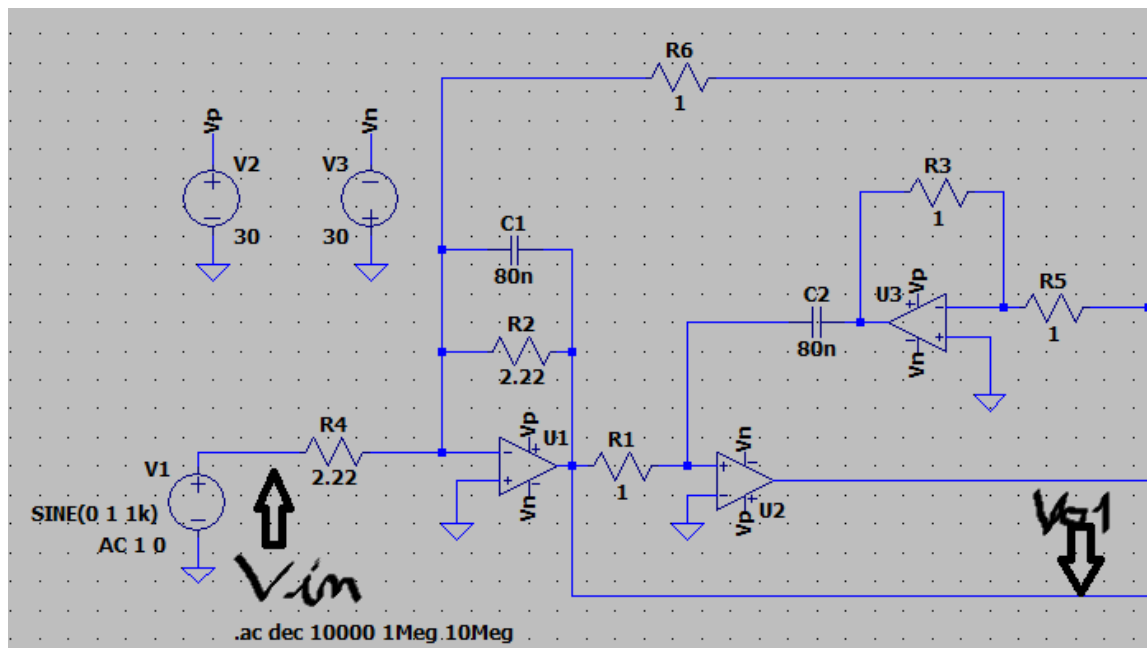
$\text{Denom}(s) = s^2 + \$0,134178 + \$1,2085 + \$0,134178 - \$1,2085 + (0,134178)^2 + (1,2085)^2$
 $\text{Denom}(s) = s^2 + \$0,268356 + 1,47726s$

Por criterios de diseño $\Rightarrow R_N = 1 \Omega = R_A = R_N$, $Z_N = 1$
 y por normalización del circuito de Schubert $\Rightarrow \left(\frac{R}{K}\right)_N = \frac{1 \Omega}{0,137} = 7,27 \Omega$, $(Q \cdot R)_N = 4153 \Omega = 4,153$

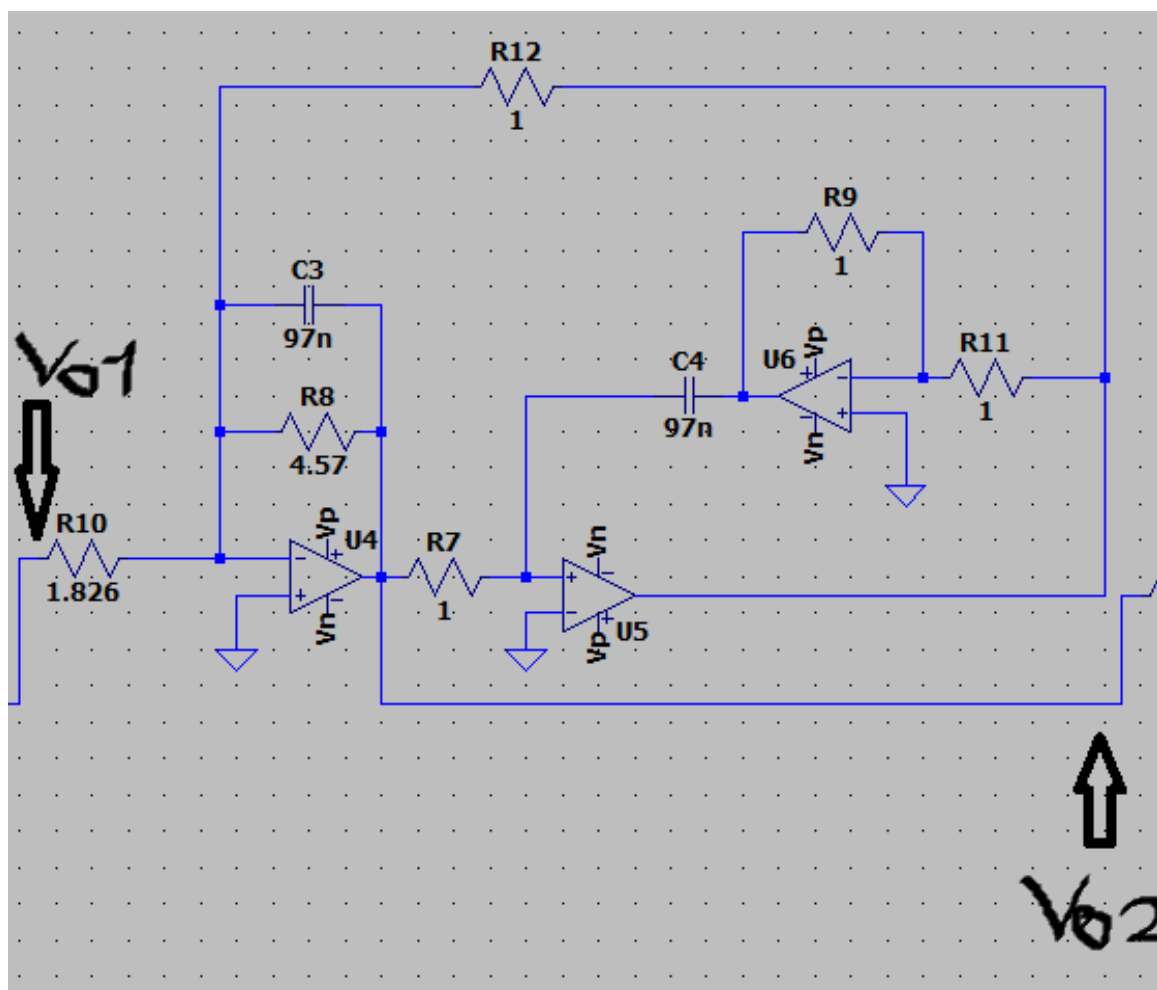
Por similitud $\Rightarrow 0,2683 = \frac{4,153}{\omega_0}$, $1,477 = \omega_0^2 \Rightarrow \omega_0 = \sqrt{1,477} = 1,2154$
 $\Rightarrow Q = \frac{1,2154}{0,2683} = 4,53$, $K \cdot \omega_0 = 0,450 \Rightarrow K = \frac{0,450}{1,2154} = 0,37$
 $\Rightarrow \omega_0 = 1 \Rightarrow \frac{1}{R_N Z_N} = 1 \Rightarrow C_N = \frac{1}{1 \Omega \cdot 1,2154} = 0,8227 \text{ F}$

Normalizando: $C = \frac{C_N}{\Omega_N Z_N}$, $R = R_N Z_N$, $\Omega_N = 12,56 \text{ Mrad/s}$

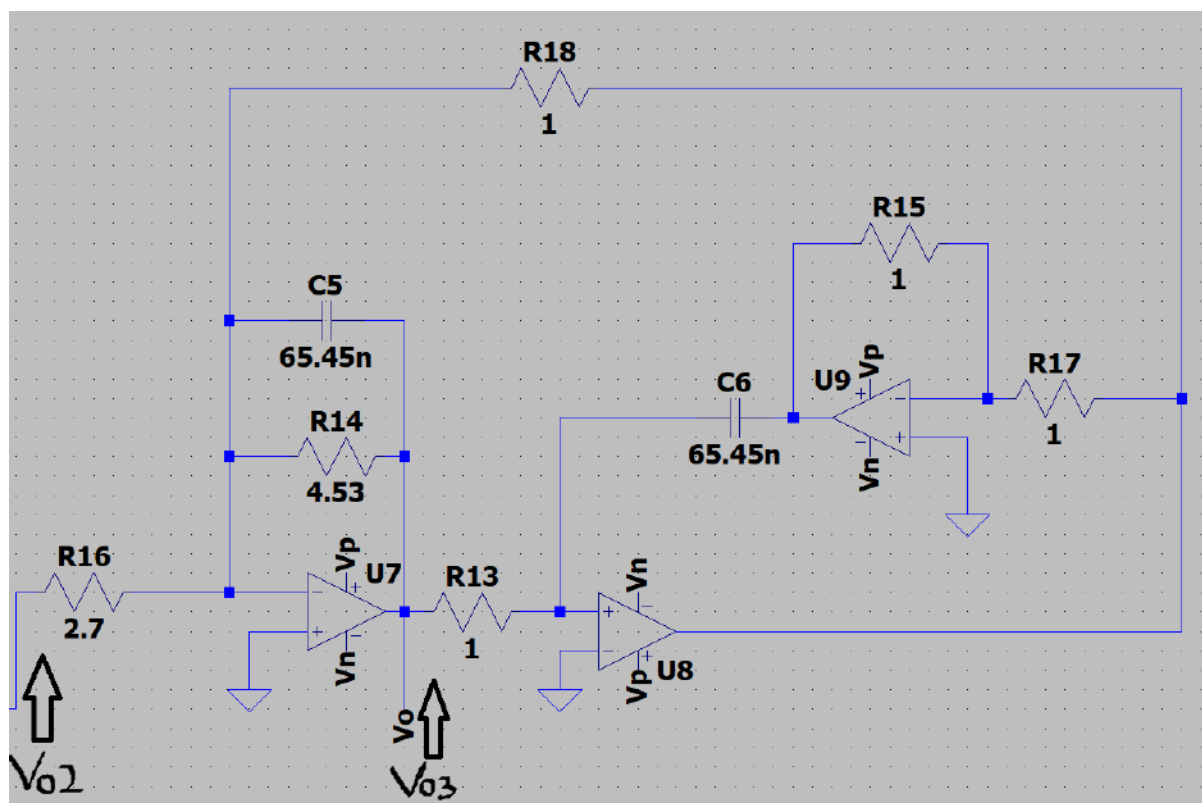
$C = \frac{0,8227 \text{ F}}{12,56 \text{ Mrad} \cdot 1} = 65,45 \text{ mF}$, $R = R_A = R_N Z_N = 1 \Omega \cdot 1 = 1 \Omega$
 $\left(\frac{R}{K}\right) = \left(\frac{R}{K}\right)_N \cdot Z_N = 2,7 \Omega \cdot 1 = 2,7 \Omega$
 $(Q \cdot R) = (Q \cdot R)_N \cdot Z_N = 4,53 \Omega \cdot 1 = 4,53 \Omega$



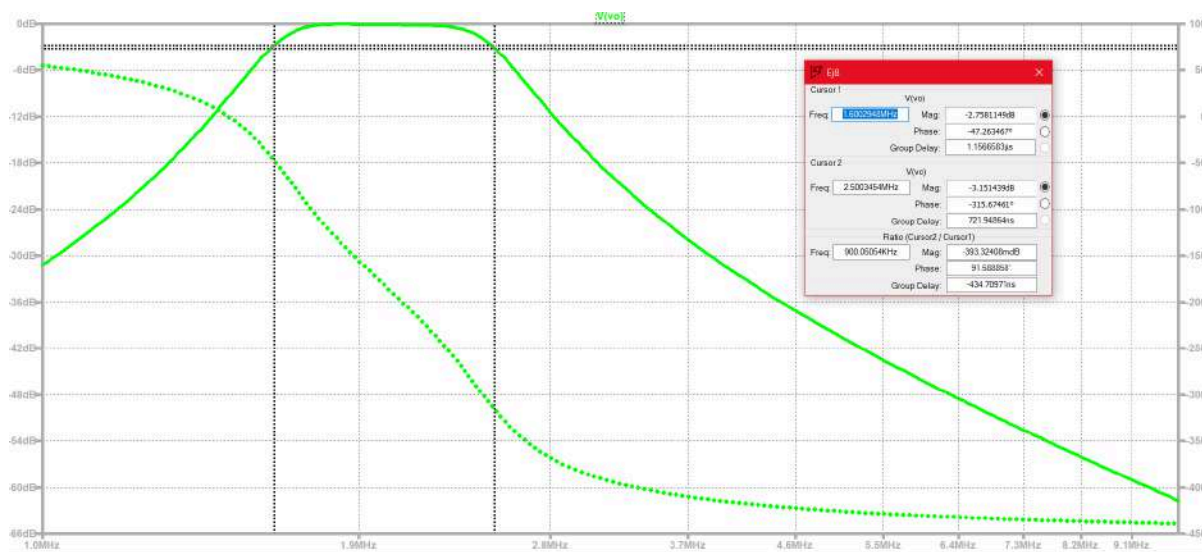
Filter circuit obtained in LTspice I.



Filter circuit obtained in LTspice II.



Filter circuit obtained in LTspice III.



Filter simulation. It is a band-pass with Maximum Flatness approximation and also Butterworth, conditions by statement. Attenuation approx. 2.75 dB at $f_{ci} = 1600$ KHz and 3.15 dB at $f_{cs} = 2500$ KHz. Of the attenuations, on average a maximum ripple of 3 dB.

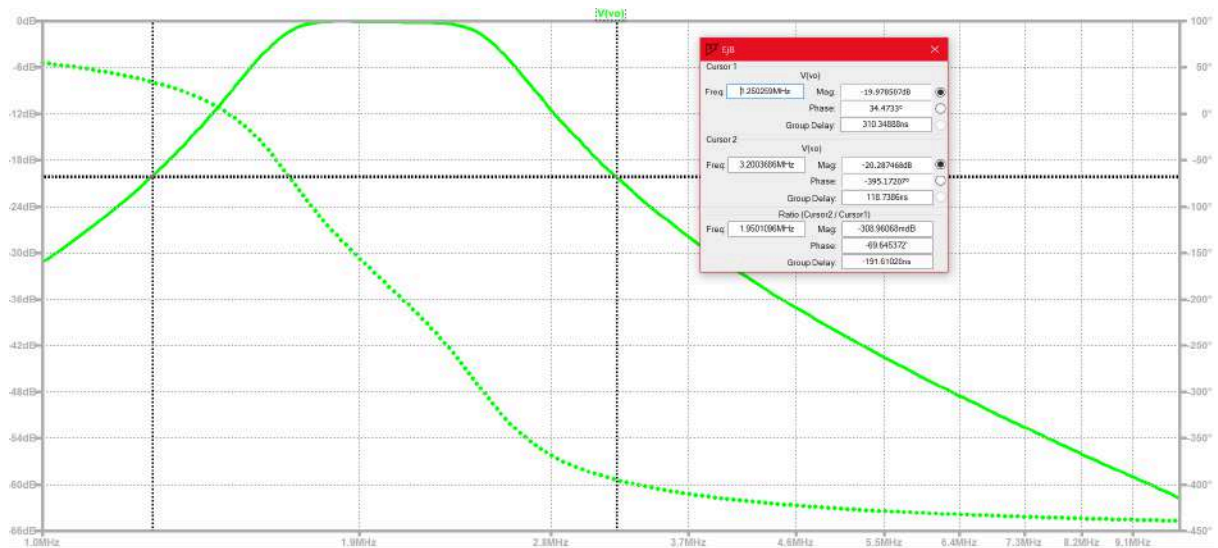


Image 21

Filter simulation. It is a band-pass with Maximum Flatness approximation and also Butterworth, conditions per statement. Attenuation approx. 20 dB at $f_{ci} = 1250$ KHz and 20.3 dB at $f_{cs} = 3200$ KHz. Of the attenuations, on average we have approximately about 20 dB for the “band stop” frequencies in forward.

Pole-Zero Diagram of $H(s)$

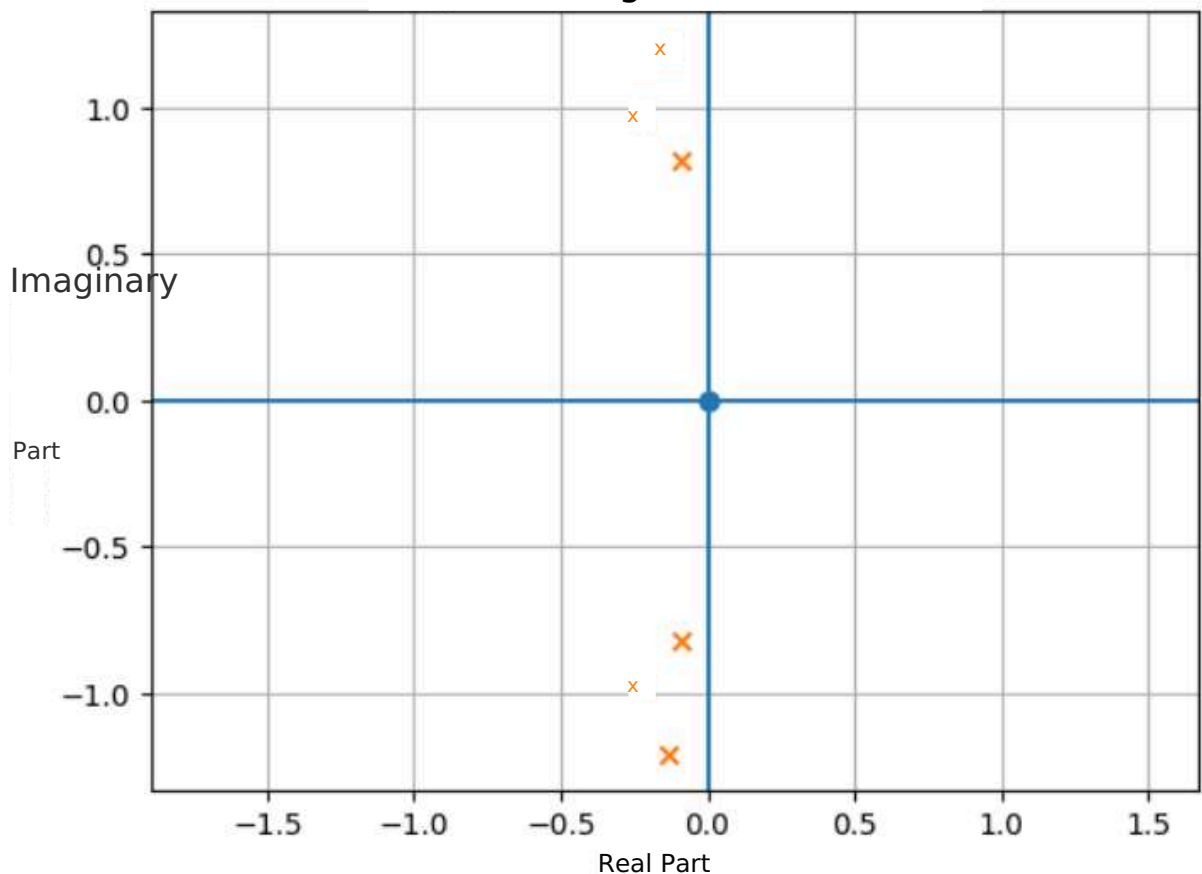


Image 22

Pole-Zero Diagram of the band-pass filter with Maximum Flatness approximation with Butterworth as a particular case.

Exercise #9

Same as previous using a Chebyshev approximation with a ripple in the passband of 1 dB

Ex. 9) Datos: $f_{p1} = 1.6 \text{ MHz}$ y $f_{p2} = 2.5 \text{ MHz}$
 - Ripple máximo en banda de paso de 1 dB, $\alpha_{\text{máx}} = 1 \text{ dB}$
 - Empíleo aproximación Chebyshev
 - Ganancia máxima en la banda de paso de 10 dB
 - Atenuación mínima de 20 dB, $\alpha_{\text{mín}} = 20 \text{ dB}$
 en las frecuencias: $f_{s1} = 1.25 \text{ MHz}$ y $f_{s2} = 3.2 \text{ MHz}$

① Como primer paso buscamos ω_0 y Q :

$$\omega_0 = \sqrt{\omega_{p1} \cdot \omega_{p2}} = \sqrt{2\pi \cdot 1.6 \text{ MHz} \cdot 2\pi \cdot 2.5 \text{ MHz}} = 12.56 \text{ MHz}$$

$$Q = \frac{\omega_0}{B} \quad \text{donde } B \text{ se calcula como: } B = \omega_{p2} - \omega_{p1}$$

$$Q = \frac{12.56 \text{ MHz}}{2\pi \cdot 2.5 \text{ MHz} - 2\pi \cdot 1.6 \text{ MHz}} = 2.22$$

② Luego, normalizo los datos de la plantilla con respecto a ω_0 .

Para normalizar con respecto a ω_0 : $\omega_N = \omega_0 \rightarrow \omega_N = 12.56 \text{ M}$

Quedando las frecuencias normalizadas así: $\omega_{0N} = \omega_0 / \omega_N = 1 \text{ Hz}$

$$\omega_{p1N} = \frac{\omega_{p1}}{\omega_N} = \frac{2\pi \cdot 1.6 \text{ MHz}}{12.56 \text{ M}} = 0.8 \text{ Hz}, \quad \omega_{p2N} = \frac{\omega_{p2}}{\omega_N} = \frac{2\pi \cdot 2.5 \text{ MHz}}{12.56 \text{ M}} = 1.25 \text{ Hz}$$

$$\omega_{s1N} = \frac{\omega_{s1}}{\omega_N} = \frac{2\pi \cdot 1.25 \text{ MHz}}{12.56 \text{ M}} = 0.625 \text{ Hz}, \quad \omega_{s2N} = \frac{\omega_{s2}}{\omega_N} = \frac{2\pi \cdot 3.2 \text{ MHz}}{12.56 \text{ M}} = 1.6 \text{ Hz}$$

• Requiriendo, utilizar el núcleo de transformación para identificar π_{SN} .

Núcleo de transformación: $\pi = f(\omega) = \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) \cdot \frac{\omega_0}{B}$

• Tenemos que $\omega_{S1N} \cdot \omega_{S2N} = 0,625 \cdot 1,6 = 1 \rightarrow$ de $f(\omega)$ podemos hallar π_{SN} y este es único:

$$\pi_{SN} = \frac{\omega_0}{B} \left(\frac{\omega_{S1} + \omega_{S2}}{\omega_0} - \frac{\omega_0}{\omega_{S1} + \omega_{S2}} \right)$$

quedando: $\pi_{SN} = \frac{12,56 \text{ MHz}}{5,6548 \text{ MHz}} \left(\frac{2\pi \cdot 1,25 \text{ MHz}}{12,56 \text{ MHz}} - \frac{12,56 \text{ MHz}}{2\pi \cdot 1,25 \text{ MHz}} \right)$

o

$$\pi_{SN} = \frac{12,56 \text{ MHz}}{5,6548 \text{ MHz}} \left(\frac{2\pi \cdot 3,2 \text{ MHz}}{12,56 \text{ MHz}} - \frac{12,56 \text{ MHz}}{2\pi \cdot 3,2 \text{ MHz}} \right)$$

dando: $\pi_{SN} = \pm 2,16 \rightarrow$ en módulo: $\pi_{SN} = 2,16$

• Calcula el n para el caso del Chebyshev:

$$n = \frac{\ln(\sqrt{4(10^{20 \cdot 0,1} - 1)} / (10^{20 \cdot 0,1} - 1))}{\ln(\pi_{SN} + \sqrt{\pi_{SN}^2 - 1})} = \frac{\ln(\sqrt{4(10^{20 \cdot 0,1} - 1)} / (10^{20 \cdot 0,1} - 1))}{\ln(2,16 + \sqrt{2,16^2 - 1})} = 2,60$$

$n = 2,60 \rightarrow n = 3$ se elige para poder cumplir con las especificaciones requeridas

• Cálculo de ϵ : $\epsilon^2 = 10^{20 \cdot 0,1} - 1 = 10^{2,0} - 1 = 0,2589$

$$\epsilon = 0,5088$$

• Luego, calculo el a (que servirá para obtener la ubicación de los polos):

$$a = \frac{1}{n} \ln \left(\frac{1}{\varepsilon} + \sqrt{\frac{1}{\varepsilon^2} + 1} \right) = \frac{1}{3} \cdot \ln \left(\frac{1}{0.508} + \sqrt{\frac{1}{0.508^2} + 1} \right) = \underline{0.476}$$

• Continuando, calculo la ubicación de los polos:

$$\Sigma_K = -\tan \left(\left(\frac{2K-1}{2n} \right) \cdot \pi \right) \cdot \tanh(a) \rightarrow K=1 \rightarrow \Sigma_{1,2} = -0.2475$$

$$K=2 \rightarrow \Sigma_3 = -0.495$$

$$r_{0K} = \cos \left(\left(\frac{2K-1}{2n} \right) \cdot \pi \right) \cdot \cosh(a) \rightarrow K=1 \rightarrow r_{0,1,2} = \pm 0.9664$$

$$K=2 \rightarrow r_{03} = 0$$

• Luego, calculo la distancia al origen de los polos:

$$r_{0K} = \sqrt{\Sigma_K^2 + r_{0K}^2} \rightarrow r_{0,1,2} = \sqrt{(-0.2475)^2 + (\pm 0.9664)^2} = 0.9975$$

$$r_{03} = \sqrt{(-0.495)^2 + (0)^2} = 0.495$$

• Calculo el Q de cada polo (solo me interesa en los complejos conjugados):

$$Q_{0K} = \frac{r_{0K}}{|\Sigma_K| \cdot 2} \rightarrow Q_{0,1,2} = \frac{r_{0,1,2}}{|\Sigma_{1,2}| \cdot 2} = \frac{0.9975}{|-0.2475| \cdot 2} = \underline{2.015}$$

a) Entonces, planteo el $H(s)$ de un paso-bajo de 3^{er} orden (que va a estar conformado por uno de 2^{do} orden y otro de 1^{er} orden):

$$H(s) = \frac{\pi_{012}^2}{s^2 + \frac{\pi_{012}}{Q_{012}} s + \pi_{012}^2} \cdot \frac{\pi_{03}}{s + \pi_{03}}$$

• Ahora tendríamos que reemplazar por el núcleo de transformación:

$$s = Q \left(\frac{W_{0N} + s}{s} \right) \quad \text{donde } W_{0N} = 1$$

• Reemplazando por el núcleo de transformación para pasar de un paso-bajos a un paso-banda, llegamos a que:

$$H(s) = \frac{\pi_{012}^2}{\left[Q \left(s + \frac{1}{s} \right) \right]^2 + Q \left(s + \frac{1}{s} \right) \frac{\pi_{012}}{Q_{012}} + \pi_{012}^2} \cdot \frac{\pi_{03}}{Q \left(s + \frac{1}{s} \right) + \pi_{03}}$$

$$H(s) = \frac{\pi_{012}^2}{Q^2 \left(\frac{s^2 + 1}{s} \right)^2 + \left(\frac{s^2 + 1}{s} \right) \frac{Q \pi_{012}}{Q_{012}} + \pi_{012}^2} \cdot \frac{\pi_{03}}{Q \left(\frac{s^2 + 1}{s} \right) + \pi_{03}}$$

$$H(s) = \frac{\pi_{012}^2}{\frac{Q^2 (s^4 + 2s^2 + 1)}{s^2} + \frac{(s^2 + 1) Q \pi_{012}}{s} + \pi_{012}^2} \cdot \frac{s^2}{s^2} \cdot \frac{\pi_{03}}{Q (s^2 + 1) + \pi_{03}} \cdot \frac{s}{s}$$

$$H(s) = \frac{s^2 \pi_{012}^2}{Q^2 (s^4 + 2s^2 + 1) + s (s^2 + 1) \frac{Q \pi_{012}}{Q_{012}} + s^2 \pi_{012}^2} \cdot \frac{s \pi_{03}}{Q (s^2 + 1) + \pi_{03}}$$

$$H(s) = \frac{s^2 \pi_{012}^2}{s^4 Q^2 + 2s^2 Q^2 + Q^2 + s^3 \frac{Q \pi_{012}}{Q_{012}} + s^2 \pi_{012}^2 + s \frac{Q \pi_{012}}{Q_{012}}} \cdot \frac{s \pi_{03}}{s^2 Q + s \pi_{03} + Q}$$

$$H(s) = \frac{s^2 \pi_{012}^2}{s^4 Q^2 + s^3 \frac{Q \pi_{012}}{Q_{012}} + s^2 (2Q^2 + \pi_{012}^2) + s \frac{Q \pi_{012}}{Q_{012}} + Q^2 \frac{1}{Q}} \cdot \frac{1}{Q} \cdot \frac{s \pi_{02}}{Q s^2 + s \pi_{03} + Q} \cdot \frac{1}{Q}$$

$$H(s) = \frac{(s \cdot \frac{\pi_{012}}{Q}) \cdot (s \cdot \frac{\pi_{012}}{Q})}{s^4 + s^3 \frac{\pi_{012}}{Q \cdot Q_{012}} + s^2 (2 + \frac{\pi_{012}^2}{Q^2}) + s \frac{\pi_{012}}{Q \cdot Q_{012}} + 1} \cdot \frac{s \frac{\pi_{03}}{Q}}{s^2 + s \frac{\pi_{03}}{Q} + 1}$$

• Reemplazo los valores en las variables:

$$H(s) = \frac{(s \cdot 0,449) (s \cdot 0,449)}{s^4 + s^3 \cdot 0,223 + s^2 \cdot 2,2019 + s \cdot 0,223 + 1} \cdot \frac{s \cdot 0,223}{s^2 + s \cdot 0,223 + 1}$$

→ A

• Los raíces del denominador A son:

$$X_{1,2} = -0,067623 \pm j1,239598j \text{ y } X_{3,4} = -0,043877 \pm j0,804319j$$

• Entonces, procedo a armar dos $H(s)$ distintos (uno para cada par de raíces complejas y conjugadas):

$$H_1(s) = \frac{s \cdot 0,449}{(s - (-0,067623 + j1,239598j))(s - (-0,067623 - j1,239598j))}$$

$$H_1(s) = \frac{s \cdot 0,449}{s^2 - s(-0,067623 - j1,239598j) - s(-0,067623 + j1,239598j) + 1,54117}$$

$$H_1(s) = \frac{s \cdot 0,449}{s^2 + s \cdot 0,135246 + 1,54117}$$

$$H_2(s) = \frac{S \cdot 0,449}{(S - (-0,043877 - j0,804319))(S - (-0,043877 + j0,804319))}$$

$$H_2(s) = \frac{S \cdot 0,449}{S^2 - S(-0,043877 - j0,804319) - S(-0,043877 + j0,804319) + 0,64885}$$

$$H_2(s) = \frac{S \cdot 0,449}{S^2 + S \cdot 0,087754 + 0,64885}$$

• Escribo el último $H(s)$ que hace falta para obtener la transferencia total.

$$H_3(s) = \frac{S \cdot 0,223}{S^2 + S \cdot 0,223 + 1}$$

• Finalmente, el $H(s)$ total quedaría así: $H(s) = H_1(s) \cdot H_2(s) \cdot H_3(s)$

$$H(s) = \frac{S \cdot 0,449}{S^2 + S \cdot 0,135246 + 1,54117} \cdot \frac{S \cdot 0,449}{S^2 + S \cdot 0,087754 + 0,64885} \cdot \frac{S \cdot 0,223}{S^2 + S \cdot 0,223 + 1}$$

• b) y c) se presentan con simulación en software.

d) Pintado con estructura Ackenberg:

$$H(s)_{\text{para-banda}} = -\frac{K \cdot Q \cdot \frac{W_0}{Q} \cdot S}{S^2 + \frac{W_0}{Q} \cdot S + W_0^2}$$

• Entonces, para $H_1(s)$ despejo W_0 , Q y K de la expresión del Ackenberg:

$$W_0^2 = 1,54117 \rightarrow W_0 = 1,2414$$

$$\frac{W_0}{Q} = 0,135246 \rightarrow Q = 9,1788$$

$$K \cdot Q \cdot \frac{\omega_0}{Q} = 0,449 \rightarrow K = 0,3616$$

Por criterio de diseño $\rightarrow R_N = 1\Omega \rightarrow R_N = R_{AN}; Z_N = 1$

y por como es el circuito Ackenberg $\rightarrow \frac{R_N}{K} = \frac{1\Omega}{0,3616} = 2,76\Omega$

$$\text{y } Q \cdot R_N = 9,1788 \cdot 1\Omega = 9,1788\Omega$$

Recordando que: $\omega_0 = \frac{1}{RC} \rightarrow C_N = \frac{1}{R_N \cdot \omega_0} = \frac{1}{1\Omega \cdot 1,2414} = 0,8055F$

Desnormalizando: $C = \frac{C_N}{\Omega_N \cdot Z_N} = \frac{0,8055F}{12,56M \cdot 1} = 64,13nF$

$$R = R_N \cdot Z_N = 1\Omega \cdot 1 = 1\Omega; \frac{R}{K} = \frac{R_N}{K} \cdot Z_N = 2,76\Omega \cdot 1 = 2,76\Omega$$

$$R_A = R_{AN} \cdot Z_N = 1\Omega \cdot 1 = 1\Omega; Q \cdot R = Q \cdot R_N \cdot Z_N = 9,1788\Omega \cdot 1 = 9,1788\Omega$$

Buscando, para $H_2(s)$ despejando ω_0 , Q y K de la expresión del Ackenberg:

$$\omega_0^2 = 0,64885 \rightarrow \omega_0 = 0,8055$$

$$\frac{\omega_0}{Q} = 0,087754 \rightarrow Q = 9,179$$

$$K \cdot Q \cdot \frac{\omega_0}{Q} = 0,449 \rightarrow K = 0,5571$$

Por criterios de diseño $\rightarrow R_N = 1\Omega \rightarrow R_N = R_{AN}; Z_N = 1$

y por como es el circuito Ackenberg $\rightarrow \frac{R_N}{K} = \frac{1\Omega}{0,5571} = 1,795\Omega$

$$\text{y } Q \cdot R_N = 9,179 \cdot 1\Omega = 9,179\Omega$$

Recordando que: $\omega_0 = \frac{1}{RC} \rightarrow C_N = \frac{1}{R_N \omega_0} = \frac{1}{1\pi \cdot 0.8055} = 1.2414F$

Desnormalizo: $C = \frac{1.2414F}{12.56M \cdot 1} = 98.83nF$

$R = 1\pi \cdot 1 = 1\pi$; $\frac{R}{K} = 1.795\pi \cdot 1 = 1.795\pi$
 $R_A = 1\pi \cdot 1 = 1\pi$; $Q R = 9.179\pi \cdot 1 = 9.179\pi$

Por último, para $H_3(s)$ despijs ω_0 , Q y K de la expresión del Ackerborg:

$\omega_0^2 = 1 \rightarrow \omega_0 = 1$

$\frac{\omega_0}{Q} = 0.223 \rightarrow Q = 4.4843$

$\frac{K \cdot Q \cdot \omega_0}{Q} = 0.223 \rightarrow K = 0.223$

Por criterio de diseño $\rightarrow R_N = 1\pi \rightarrow R_{AN} = R_N$, $Z_N = 1$

y por como es el circuito Ackerborg $\rightarrow \frac{R_N}{K} = \frac{1\pi}{0.223} = 4.4843\pi$

y $Q \cdot R_N = 4.4843 \cdot 1\pi = 4.4843\pi$

Recordando que: $\omega_0 = \frac{1}{RC} \rightarrow C_N = \frac{1}{R_N \omega_0} = \frac{1}{1\pi \cdot 1} = 1F$

Desnormalizo: $C = \frac{1F}{12.56M \cdot 1} = 79.61nF$

$R = 1\pi \cdot 1 = 1\pi$; $\frac{R}{K} = 4.4843\pi \cdot 1 = 4.4843\pi$
 $R_A = 1\pi \cdot 1 = 1\pi$; $Q R = 4.4843\pi \cdot 1 = 4.4843\pi$

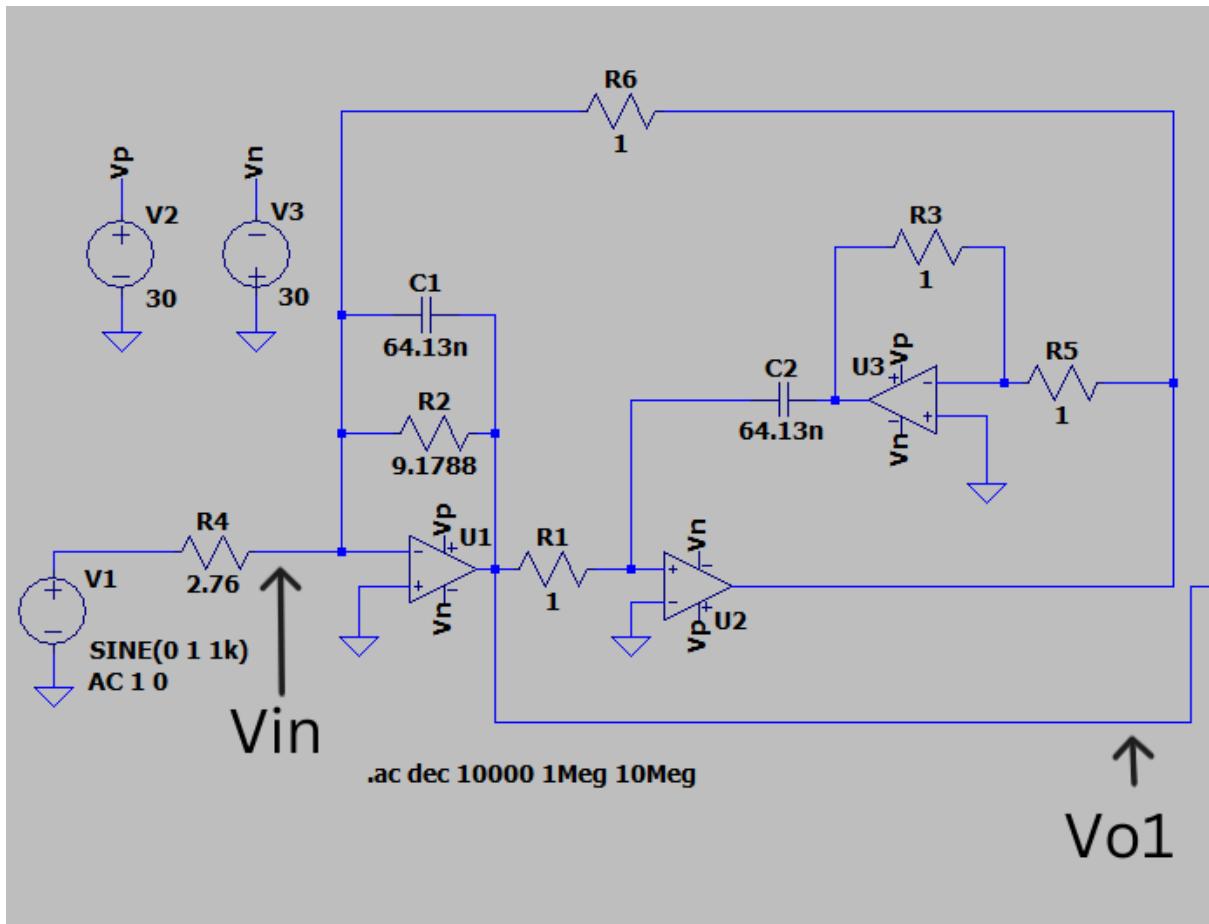


Imagen 23

Filter circuit obtained in LTspice I.

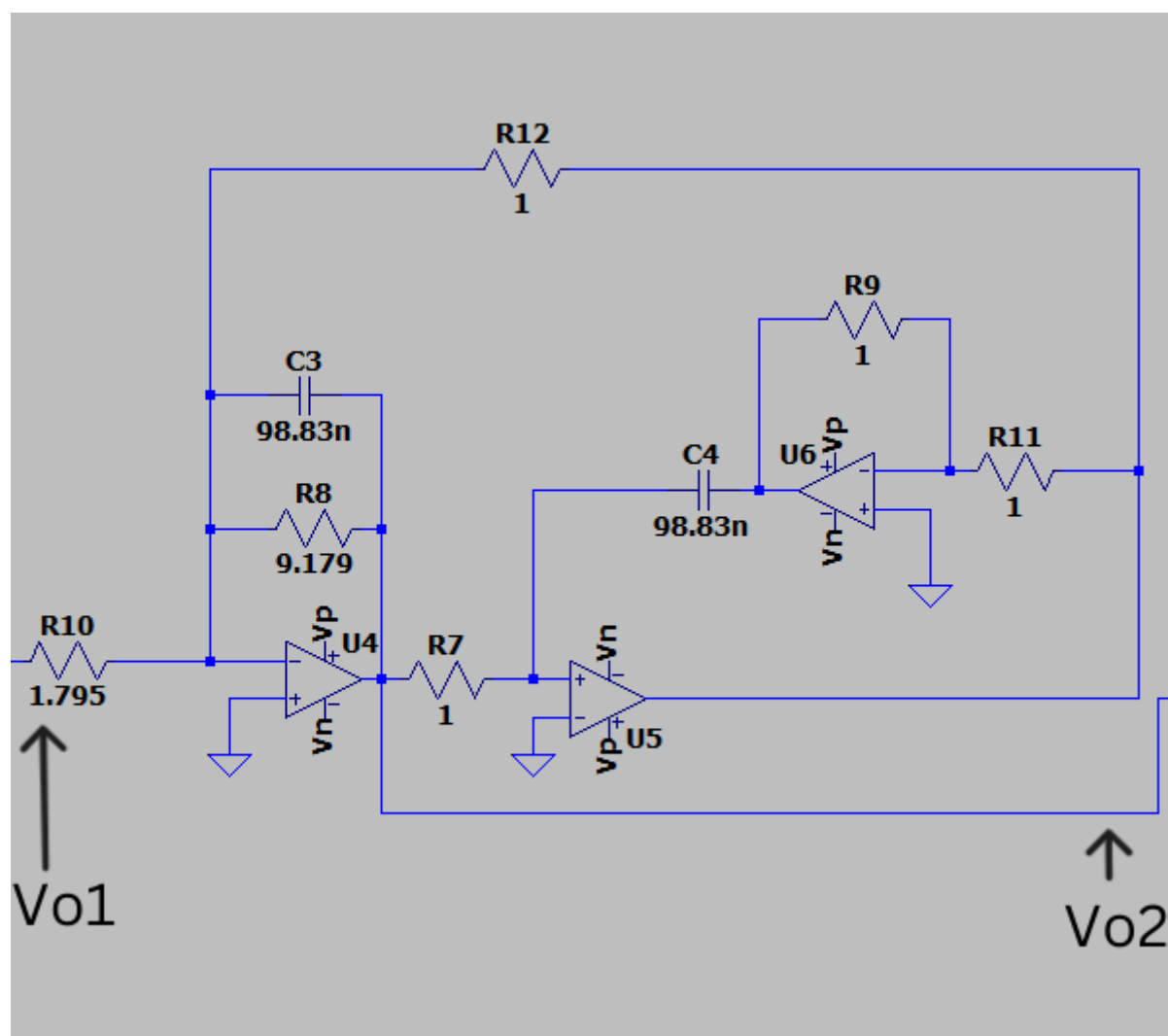


Imagen 24

Filter circuit obtained in LTSpice II.

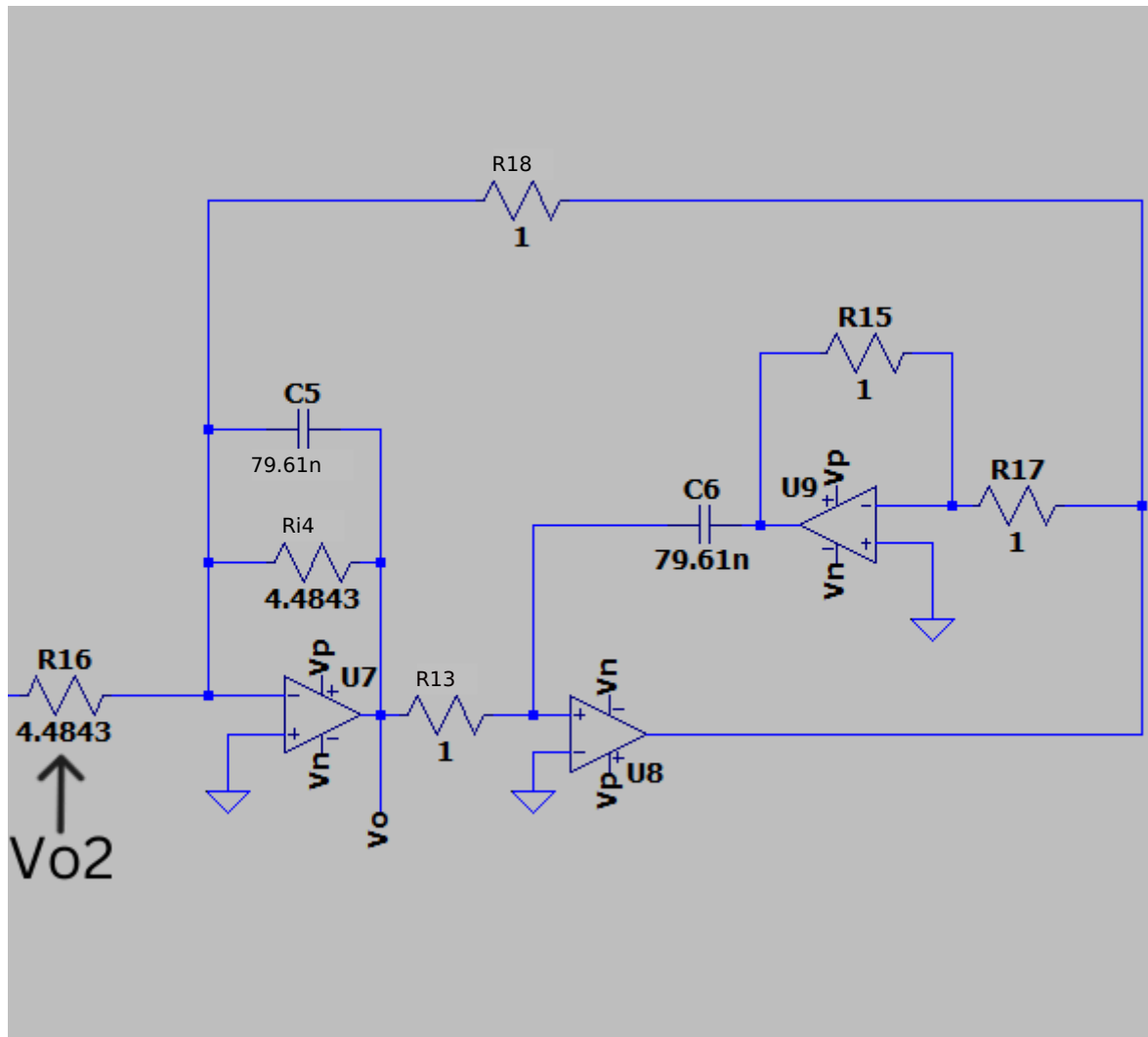


Image 25

Circuit of the filter obtained in LTspice III.

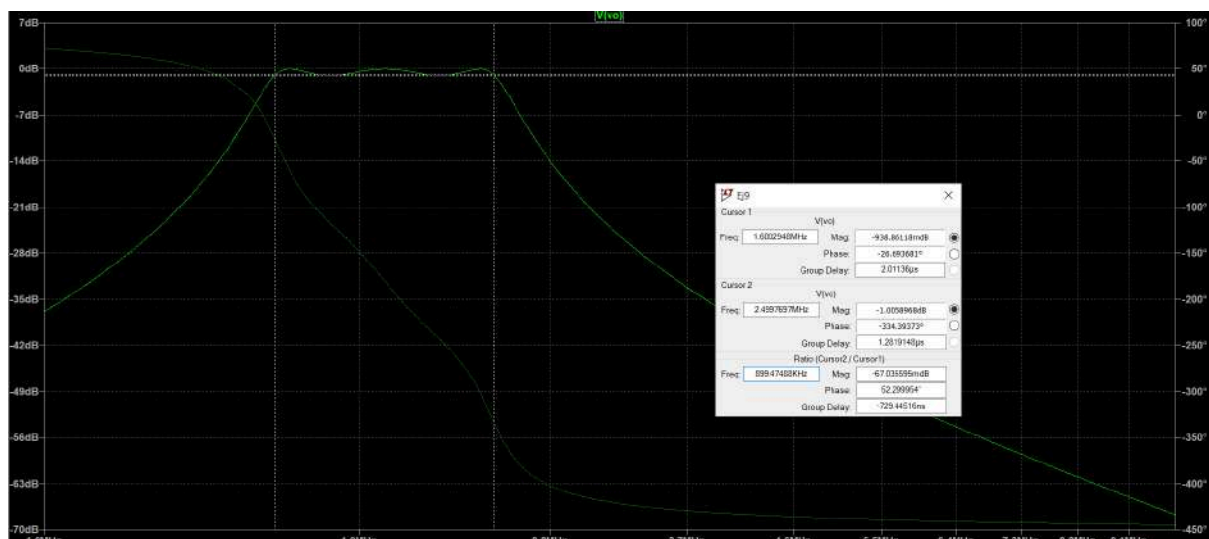


Imagen 26

Filter simulation. It is a band-pass with Chebyshev approximation, conditions by the statement. Attenuation approx. 0.94 dB at fci = 1600 KHz and 1 dB at fcs = 2500 KHz. Of the attenuations, on average a Ripple maximum of 1 dB.

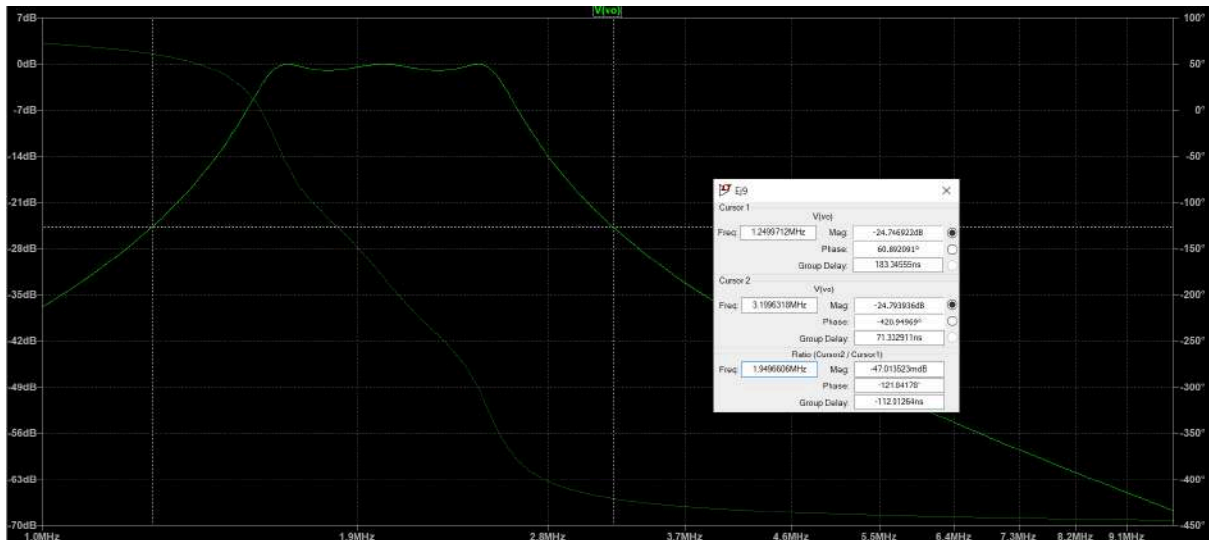


Image 27

Filter simulation. It is a bandpass with Chebyshev approximation, conditions per statement. Attenuation approx. 24.75 dB at $f_{ci} = 1250$ KHz and 24.79 dB at $f_{cs} = 3200$ KHz. Of the attenuations, on average we have approximately about 20 dB for the frequencies of “band stop” onwards.

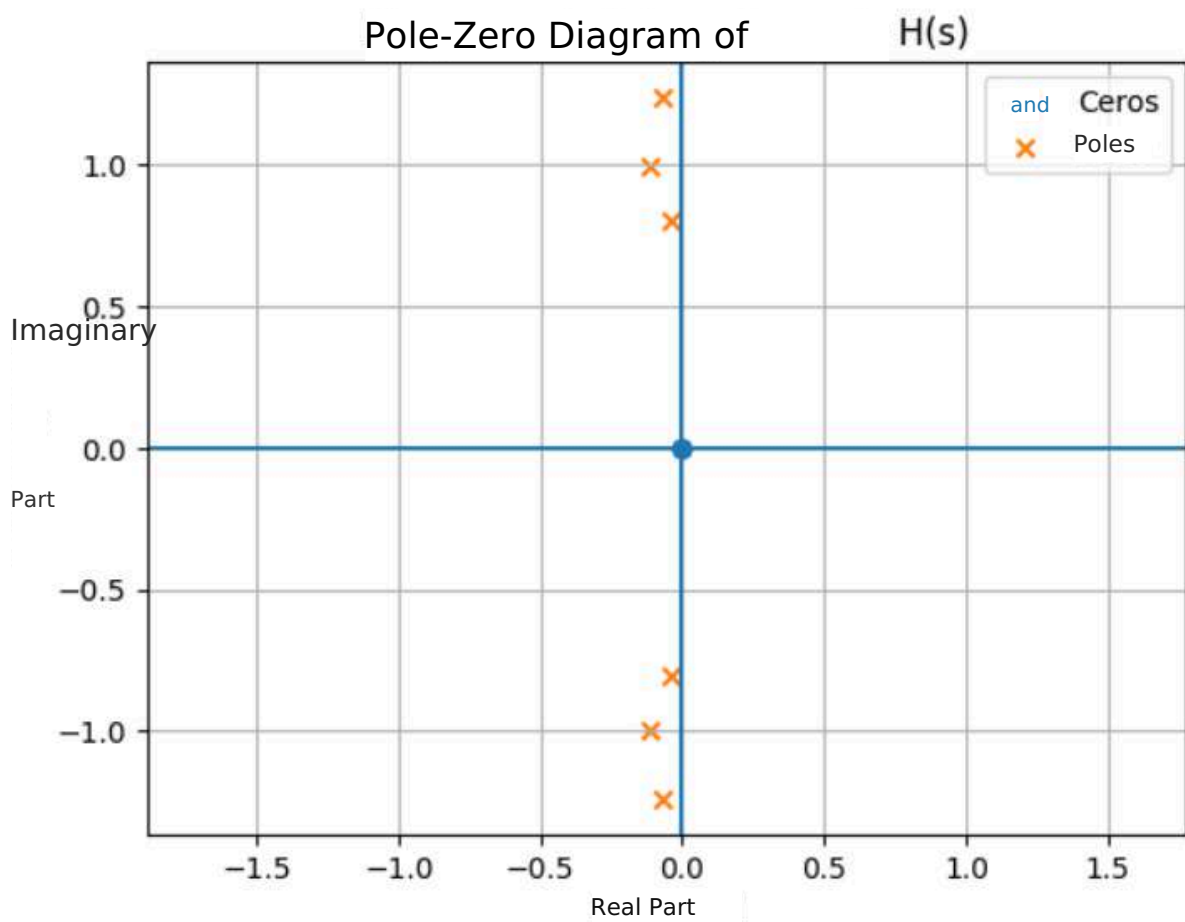


Image 28

Pole-zero diagram of the band-pass filter with Chebyshev approximation.

Exercise #10

A tone of 45 KHz and a tone of 45 KHz and 200 mV of amplitude is distorted by a tone of 12 KHz and 2 V of amplitude. Design a high-pass filter that attenuates the interfering signal, such that the remnant is not greater than 2% of the 200 mV. The gain at high frequency should be 0 dB and the maximum attenuation in the passband less than 1 dB. Use the approximation that requires the fewest stages. Synthesize the filter using Sallen-Key structures.

10) Datos: 200mV a 45KHz señal de interés
 2V a 12KHz señal de interferencia
 2% de 200mV a 12KHz señal remanente de interferencia
 Ganancia filtro = 0dB, $\text{cm} \leq 1$ en banda de paso.

1º Definir la frecuencia de corte del filtro paso alto.

Tomar un margen de seguridad y no justo 45KHz, sino permitir
 factor de interferencia $\Rightarrow 2,8\%$ de 45KHz $\approx 1250 \Rightarrow 45KHz - 1250 = f_c$
 $f_c = 43,75KHz$.

2º Definir la frecuencia de stopband del filtro paso alto.

Tomar un margen de seguridad y no justo 12KHz, sino permitir
 factor de señal interferente o lo solista $\Rightarrow 4,2\%$ de 12KHz $\approx 0,5KHz$
 $\Rightarrow 12KHz + 0,5KHz = f_e = 12,5KHz$

3º Buscar la atenuación en la banda de stop

Sabemos que queremos amplitud de señal interferente de como mucho
 2% de 200mV = 4mV $\Rightarrow$ por lo tanto una atenuación en la
 banda de stop tal que formos de 2V $\Rightarrow$ 4mV de amplitud remanente luego
 del filtro, entonces e_{rel}

$$-20 \log \left(\frac{4mV}{2V} \right) = -20 \log \left(\frac{V_{interferencia}}{V_{señal}} \right) \approx 55dB = \text{atenuación}$$

Atenuación = Es la atenuación mínima de paso que se le da a la señal interferente
 no permitiendo luego del filtro, sea la más pequeña posible.

4º Normalizar respecto de la potencia de salida, $P_{out,HP} = P_{in} = \frac{P_{HP}}{P_{HP}}$
 $\Omega_N = 2\pi N \Delta = 2\pi \cdot 4375 \times 10^3$ And $\Omega_{HP} = 1 = \frac{\omega_c}{\Omega_N} = \frac{1}{3.5} = 0.2857$
 $N \Delta = f_c (\text{normalizado}) = 4375 \times 10^3$ $\Omega_S = 3.5 = \frac{\omega_s}{\Omega_N} = \frac{2\pi \cdot 12.5 \times 10^3}{2\pi \cdot 4375 \times 10^3} = \frac{2}{7} = \frac{1}{3.5} = 0.2857$

5º Elip método de optimización y luego el "m" de E

Como se pide el menor orden "m" posible $\rightarrow$ elip Chebyshev, donde

$$m = \frac{\ln \left(\sqrt{4(10^{\frac{\alpha_{\text{dén}}}{20}} - 1) / (10^{\frac{\alpha_{\text{num}}}{20}} - 1)} \right)}{\ln \left(\Omega_S + \sqrt{\Omega_S^2 - 1} \right)} \quad \Omega_S = \Omega_{\text{stop}} = 3.5 \quad \alpha_{\text{num}} = 55 \text{ dB}, \alpha_{\text{dén}} = 40 \text{ dB}$$

$$m = \frac{\ln \left(\sqrt{4(10^{5.5} - 1) / (10^{0.1} - 1)} \right)}{\ln \left(3.5 + \sqrt{3.5^2 - 1} \right)} = 4.00765 \approx 4$$

$$\epsilon^2 = 10^{\frac{\alpha_{\text{dén}}}{20}} - 1 = 10^{0.1} - 1 = 0.258925 = \epsilon^2 \Rightarrow \epsilon = 0.5088$$

6º Alternar $H(s)$ Low Pass y $H(s)$ High Pass

$$H(s)_{\text{Low Pass}} = \frac{\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2} \quad \text{orden } m=2, \quad H(s)_{\text{High Pass}} = \frac{\omega_0}{s + \omega_0} \quad \text{orden } m=1$$

y reemplazando s por $\frac{1}{s}$:

Ara $H(s)_{\text{High Pass}} = H(s)_{\text{Low Pass}} \Big|_{s=\frac{1}{s}} = \frac{\omega_0^2}{\left(\frac{1}{s}\right)^2 + \frac{\omega_0}{Q} \cdot \frac{1}{s} + \omega_0^2} \cdot \frac{s^2 \omega_0^2}{1 + \frac{\omega_0}{Q}s + \omega_0^2 s^2} \quad (\text{fin})$

Ue $H(s)_{\text{High Pass}} = \frac{s^2}{\frac{1}{\omega_0^2} + \frac{1}{Q\omega_0}s + s^2} \quad \text{orden } m=2$

$$H(s)_{\text{High Pass}} = H(s)_{\text{Low Pass}} \Big|_{s=\frac{1}{s}} = \frac{\omega_0}{\frac{1}{s} + \omega_0} \cdot \frac{s}{s} = \frac{\omega_0 \cdot s}{1 + \omega_0 \cdot s} \cdot \frac{1/\omega_0}{1/\omega_0}$$

$$H(s)_{\text{High Pass}} = \frac{s}{\omega_0 + s} \quad \text{orden } m=1$$

Nota: Por cada transpuesta de orden "2" por los polos, relacionamos otro transpuesta de orden "2" por los ceros.

7º Aplicar diversas fórmulas de Shlyshevsky para encontrar los polos y los $W_0 = \Omega_0 K$ de cada fondo de polos.

$$z = \frac{1}{m} \ln \left(\frac{1}{E} + \sqrt{\frac{1}{E^2} + 1} \right), \quad \sigma_K = \operatorname{sn} \left(\frac{(2K-1)\pi}{2m} \right) \operatorname{sn}(z), \quad \Omega_K = \cos \left(\frac{(2K-1)\pi}{2m} \right) \cos(z)$$

$$\Omega_0 K = \sqrt{\sigma_K^2 + \Omega_K^2}, \quad \sigma_K = \operatorname{arctg} \left(\frac{\Omega_K}{\sigma_K} \right), \quad \frac{\sigma_K^2}{\operatorname{sn}^2(z)} + \frac{\Omega_K^2}{\cos^2(z)} = 1$$

$$\operatorname{sn}(z) = \frac{e^z - e^{-z}}{2}, \quad \cos(z) = \frac{e^z + e^{-z}}{2}$$

$$\sigma_K = \frac{\Omega_0 K}{\Omega_K}$$

Transformación geométrica $z \rightarrow w$

$$m=4, \quad E^2 = 0.258925, \quad E = 0.5088, \quad \sigma = 0.354$$

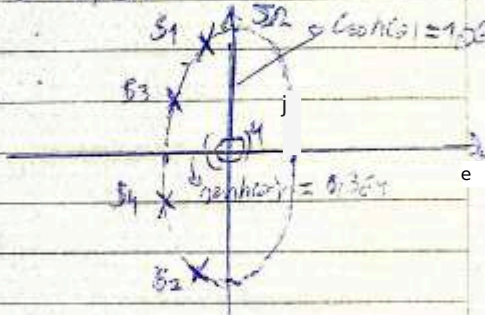
$$\sigma_{1,2} = -0.14, \quad \sigma_{3,4} = 0.3368$$

$$\Omega_{1,2} = \pm 0.983, \quad \Omega_{3,4} = \pm 0.4073$$

$$\Omega_{0,1,2} = 0.9729 \times 1, \quad \Omega_{0,3,4} = 0.5285$$

$$\theta_{1,2} = \pm 8.189^\circ, \quad \theta_{3,4} = \pm 50.41^\circ$$

$$\rho_{0,1,2} = 3.5714, \quad \rho_{0,3,4} = 0.7845$$

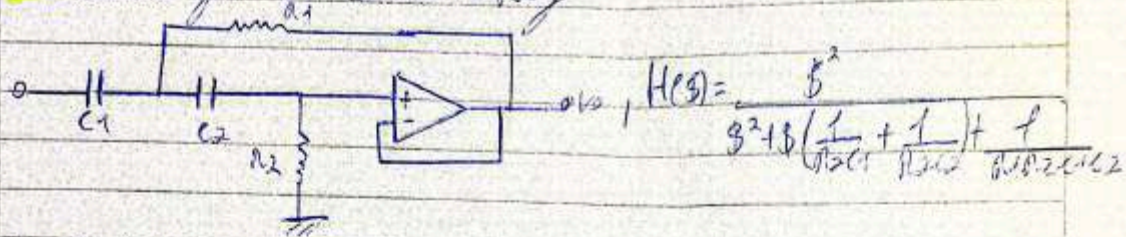


$$H(s)_{\text{poro Alter}} = H_1(s)_{\text{poro Alter}} \cdot H_2(s)_{\text{poro Alter}}$$

$$H_1(s)_{\text{poro Alter}} = \frac{s^2}{1 + 3.928s + s^2} = \frac{s^2}{[s - (-0.14 - 0.983j)][s - (-0.14 + 0.983j)]}$$

$$H_2(s)_{\text{poro Alter}} = \frac{s^2}{3.58 + 2.492s + s^2} = \frac{s^2}{[s - (0.3368 + 0.4073j)][s - (0.3368 - 0.4073j)]}$$

3º Sintetizar can Sallen-Key



$$H(s) = \frac{s^2}{s^2 + s \left(\frac{1}{R_2 C_1} + \frac{1}{R_1 C_2} \right) + \frac{1}{R_1 R_2 C_1 C_2}}$$

1ª etapa: $C_{1,2} = 1\text{F} \wedge Z_N = 1\text{K}$ for external de diseño

$$Q_{2,1} = \frac{1}{R_{2N} C_{1N}} + \frac{1}{R_{1N} C_{2N}} \Rightarrow R_{2N} = 7,14\text{K}$$

$$1 = \frac{1}{R_{1N} R_{2N} C_{1N} C_{2N}} \Rightarrow R_{1N} = 0,14\text{K} = 140\Omega$$

Dinamizaje: $C = \ln \frac{R}{s_N \cdot Z_N}$, $R = R_N \cdot Z_N$, $R_N = 2\pi \cdot 43,75\text{KHz}$

$$C_{1,2} = \frac{1\text{F}}{2\pi \cdot 43,75\text{K} \cdot 1\text{K}} = 3,638\text{mF}$$

$$R_1 = 0,14\text{K} = 140\Omega, R_2 = 7,14\text{K} = 7140\Omega$$

2da etapa: $C'_{1,2} = 1\text{F} \wedge Z_N = 1\text{K}$ for external de diseño

$$Q_{2,1}' = \frac{1}{R_{2N}' C_{1N}'} + \frac{1}{R_{1N}' C_{2N}'} \Rightarrow R_{2N}' = 0,83\text{K}$$

$$Q_{1,2}' = \frac{1}{R_{1N}' R_{2N}' C_{1N}' C_{2N}'} \Rightarrow M_V: 0,3365\text{K}$$

Dinamizaje: $C' = \frac{C_N'}{s_N \cdot Z_N}$, $R = R_N' \cdot Z_N$, $R_N' = 2\pi \cdot 43,75\text{KHz}$

$$C'_{1,2} = \frac{1\text{F}}{2\pi \cdot 43,75\text{K} \cdot 1\text{K}} = 3,638\text{mF}$$

$$R_1' = 0,3365\text{K} = 336,5\Omega, R_2' = 0,83\text{K} = 830\Omega$$

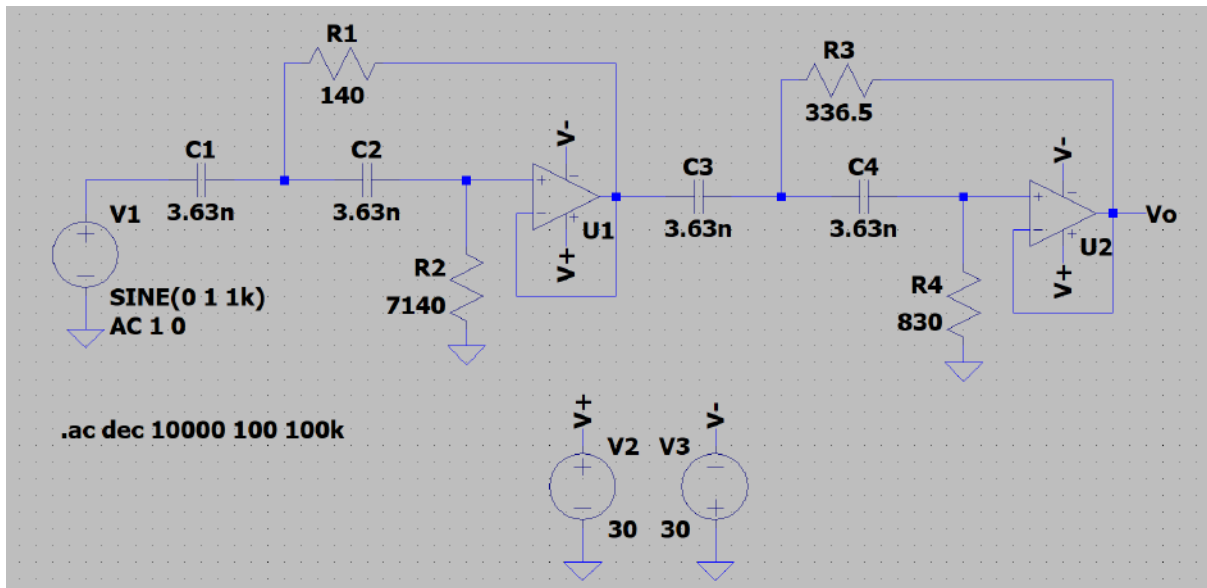
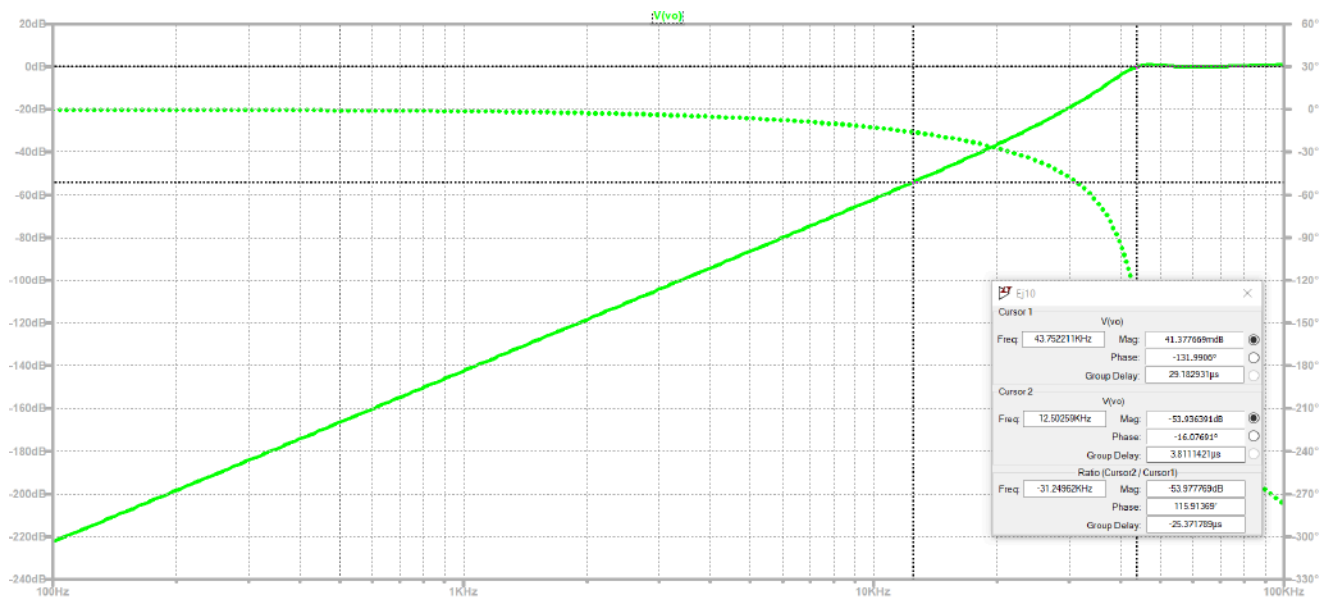


Imagen 29

Filter circuit obtained in LTspice.



Filter simulation. It is a high-pass filter with Chebyshev approximation, conditions as stated. Approximate attenuation of 0.41 dB at $f_c = 43.75$ KHz and 53.93 dB at $f_s = 12.5$ KHz. From the attenuations, on average we have approximately 54 dB of attenuation between one frequency and another.

Imagen 30